Barra China A Total Market Equity Model for Long-Term Investors

Empirical Notes

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August 2018

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1 Introduction

This document provides a methodology overview and the empirical results for MSCI's new Barra China A Total Market Equity Model for Long-Term Investors[1]. These notes include the descriptions of key modeling decisions and methodologies, extensive information on the factor structure, an analysis of the explanatory power of the model, and a summary of the factors statistics. Furthermore, these notes include a thorough side-by-side comparison of the forecasting accuracy and backtest performance of the new model and its predecessor, the CNE5 Model. The new model incorporates many of the latest innovations by MSCI used for building multi-factor equity models, including new Systematic Equity Strategies (SES) factors, latest advances in descriptor research, and the alignment of factor structure with investment horizon.

The new style factors included in the Barra China A Long-Term Model significantly enhance the information content of the model. The CNE5 Model includes three SES factors, i.e., Book-to-Price, Earnings Yield, and Momentum. The Long-Term Model includes five new SES factors, i.e., Dividend Yield, Earnings Quality, Investment Quality, Long-Term Reversal, and Profitability. In addition, the new Long-Term Model also includes the Earnings Variability factor to capture the uncertainty in earnings. These new factors, together with the significant data and methodological enhancements made to other existing factors, improve the accuracy of risk forecasts by the model, especially for the portfolios that have non-trivial tilts toward these factors. The addition of the new SES factors, in particular, helps to monitor exposures to potentially crowded trades.

The alignment of factor structure with investment horizon is one of the key innovations of new Barra equity models. The suite of Barra China A Total Market Equity Models[2] consists of

- Barra China A Total Market Equity Model for Long-Term Investors, and
- Barra China A Total Market Equity Trading Model (*coming soon*)

The Long-Term Model is comprised of stable factors. These factors tend to be the focus of long-term investments with low portfolio turnover. By contrast, the Trading Model includes additional factors that are relevant at shorter investment horizons but are considered to be too fast, that is, exposures of the assets to these factors change too much and too quickly, for long-term investors. The Long-Term Model helps achieve risk-reduction while keeping portfolio turnover and transaction costs at a low level.

In summary, the Barra China A Long-Term Model offers the following main features[3]:

- Alignment of the factor structure with a long-term investment horizon for the reduction of portfolio turnover
- Additional Systematic Equity Strategies to capture new sources of investment risk and detect potentially crowded trades
- Point-in-time fundamental data to facilitate more realistic backtests

[1] We also refer to this model as the Barra China A Long-Term Model or the Long-Term Model in this document.

[2] We also refer to the suite of Barra China A Total Market Equity Models as the Barra China A Equity Models.

[3] As other recently-introduced equity models, the Barra China A Long-Term Model also applies Volatility Regime Adjustment for factor and specific risk, and Optimization Bias Adjustment for factor covariance.

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- Daily updates of the model
- 32 industry factors based on the Global Industry Classification Standard (GICS®)

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2 Methodology Highlights

2.1 Factor Structure and Investment Horizon

The new Barra China A Equity Models suite includes the Long-Term Model and the Trading Model. The factor sets of the two models are aligned with their respective investment horizons: the Long-Term Model is intended for a monthly horizon or longer and the Trading Model is intended for a much shorter horizon, typically in the range of a few days. The Long-Term Model incorporates stable risk factors that are effective in capturing long-term stock volatilities and correlations. The Trading Model includes additional factors that change much faster.

We use factor exposure stability, which we define as the cross-sectional correlation of factor exposures across a given time period, to assess the relevance of a factor for the time period. Factor exposure stability is a measure of the rate of information decay. It depends on both the frequency of data updates and the magnitude of the changes between the updates. A high frequency of data updates and large changes between the updates tend to result in faster information decay and less stability in factor exposures.

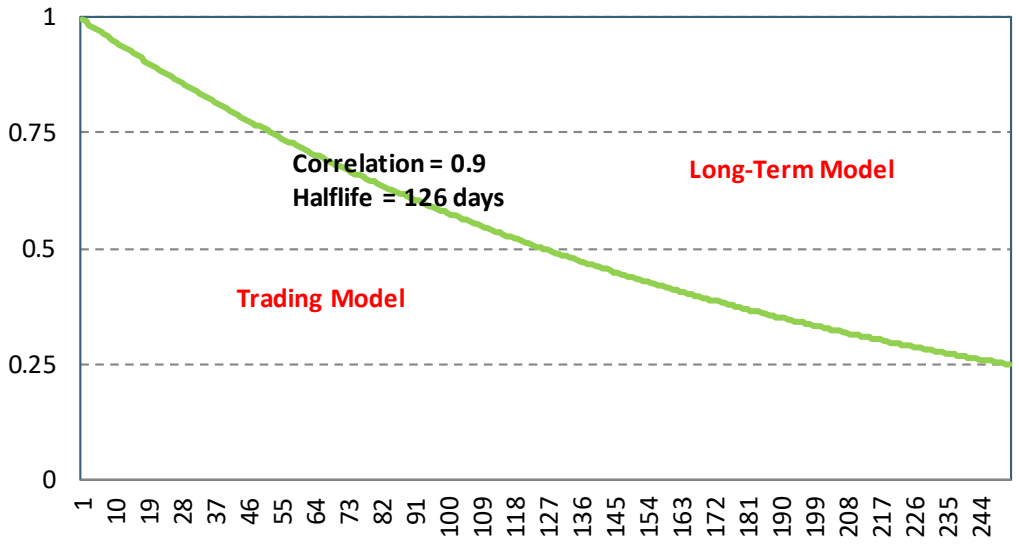
There is a direct link between factor exposure stability and portfolio turnover. For faster factors, the changes in underlying information is faster and larger, implying that portfolio positions need to be rebalanced more frequently and more substantially to reflect the updates in factor exposures optimally. This will lead to a higher portfolio turnover. While a high turnover may be required to pursue short-term strategies, it may be detrimental to the performance of long-term investors due to transaction costs.

In Figure 2.1, we depict how the factor exposure stability decays with time for a factor with a monthly stability of 0.9, which is equivalent to a 126-day half-life. Factors that are more stable will have decay curves above this line, and are included in the Long-Term Model. Factors that fall below the line are only included in the Trading Model.

By keeping only stable factors, the Barra China A Long-Term Model mitigates the risk sources that disappear over the long-term investment horizon, and reduces turnover in constructing portfolios. Therefore, the Long-Term Model is better suited for typical use cases of long-term investors.

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Figure 2.1: Factor Stability



2.2 Systematic Equity Strategies as Risk Factors

The concept of Systematic Equity Strategies was introduced and discussed by Bayraktar et al (2013) and is implemented in the more recent Barra Equity Models. The following eight strategies are included in the Barra China A Long-Term Model as style factors:

- Book-to-Price
- Dividend Yield
- Earnings Quality
- Earnings Yield
- Investment Quality
- Long-Term Reversal
- Momentum
- Profitability

Book-to-Price, Earnings Yield, and Momentum are available in the predecessor model, CNE5. The other five factors are additions to the Barra China A Long-Term Model.

These factors are commonly employed by practitioners either as factors in their quantitative process or as screens for fundamental investment. By including these factors, the Barra China A Long-Term Model allows investors to measure their exposure to popular, but potentially crowded investment strategies. Furthermore, asset managers can attribute risk and return of a portfolio to these factors and obtain more meaningful insights into drivers of their investment strategies.

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The empirical analysis in this paper supports our intuition that the inclusion of Systematic Equity Strategy factors in a risk model can lead to more accurate risk forecast and enhanced portfolio performance, particularly for portfolios that are based on a systematic investment approach.

2.3 Point-in-Time Fundamental Data

Companies often make adjustments to selected items in their past financial statements, and may even restate their financial statements for some fiscal periods entirely. In backtests, it is desirable for portfolio managers to be able to replicate a descriptor with the data that was available at the time referenced in the test. Such a capability is known as point-in-time.

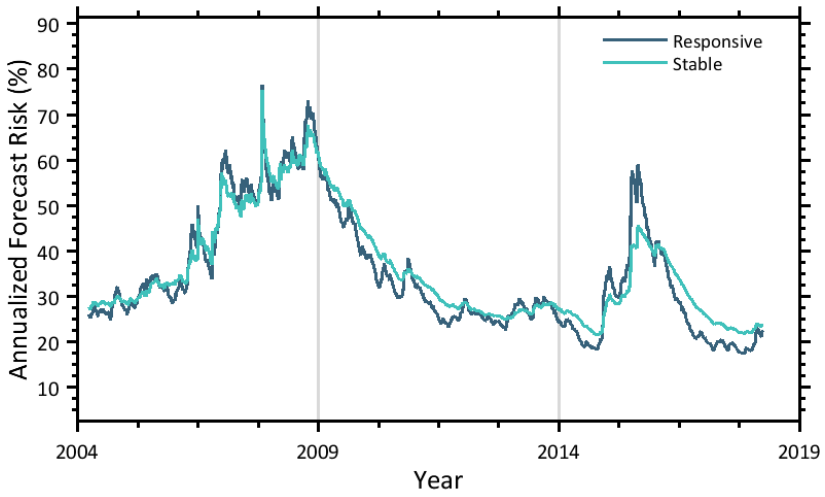
The new Barra China A Equity Models support this capability by storing the fundamental data with not only its reporting fiscal period, but also its actual publication date. When constructing the descriptors for a given date, the availability of the publication date allows us to use only the data published before that date, thus avoiding a possible look-ahead bias.

2.4 Model Responsiveness

The Barra China A Long-Term Model comes in two variants: Responsive and Stable. Both variants use the same factor structure and factor returns, but differ in model estimation parameters. The Responsive variant uses a shorter half-life to estimate factor and specific volatilities, therefore adapts faster to changes in these volatilities. The Stable variant uses a longer half-life and therefore is less responsive.

In Figure 2.2, we plot the time series of volatility forecasts for the market capitalization-weighted estimation universe given by both variants. We see that the Responsive variant adapts much faster to market shocks.

Figure 2.2: Volatility forecasts for the cap-weighted estimation universe of the Barra China A Long-Term Model



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2.5 Volatility Regime Adjustment

A major source of bias in a risk model is the change in the level of volatility, a characteristic known as non-stationarity. Because risk models look backward to make predictions about the future, they tend to under-predict risk in times of rising volatility and over-predict risk in times of falling volatility.

The Volatility Regime Adjustment is used for adjusting factor volatilities. The method is described in detail by Menchero and Morozov (2015), where it is empirically compared with simple volatility forecasts for the US equity market and factors. It relies on a cross-sectional bias statistic, which may be interpreted as an instantaneous measure of risk model bias. By taking a weighted average of this measure over a suitable interval, the bias can be significantly reduced.

Just as factor volatilities are not stable across time, the same holds true for specific risk. We therefore also apply Volatility Regime Adjustment to the specific risk model.

For more technical details on Volatility Regime Adjustment, see Appendix A.

2.6 Optimization Bias Adjustment

Another significant bias exhibited by risk models is the tendency to under-predict the risk of optimized portfolios, as demonstrated empirically by Muller (1993). Bender et al (2009) derived an analytic result for the magnitude of the bias, showing that the under-forecasting becomes increasingly severe as the number of factors grows relative to the number of time periods used to estimate the factor covariance matrix. The basic source of this bias is sampling error. Specifically, spurious correlations may cause certain stocks to appear as good hedges in-sample, but these hedges fail to perform as effectively out-of-sample.

We identify portfolios that capture biases and correct them directly within the factor covariance matrix. As shown by Menchero, Wang, and Orr (2011), the eigenfactors of the sample covariance matrix are systematically biased. Specifically, the sample covariance matrix tends to under-predict the risk of low-volatility eigenfactors, while over-predicting the risk of high-volatility eigenfactors.

Similar to the CNE5 Model, we estimate the biases via Monte Carlo simulation and then adjust the eigenvalues of the estimated covariance matrix to correct these biases. This procedure helps improve factor risk forecasts for optimized portfolios. Furthermore, it builds the corrections directly into the factor covariance matrix, while fully preserving the meaning and intuition of the pure factors. Lee et al (2011) demonstrate the effectiveness of this approach by backtesting active portfolios.

For more technical details on Optimization Bias Adjustment, see Appendix B.

2.7 Specific Risk Model with Bayesian Shrinkage

The specific risk model builds upon methodological advances introduced with the latest generation of Barra Equity Models. The model estimates the specific risk directly from the time series of daily specific returns of the stocks. A significant benefit of this approach is that specific risk is estimated individually for each stock, thus reflecting the idiosyncratic nature of this risk source.

A potential shortcoming of a pure time-series approach is that specific volatilities may not fully persist out-of-sample. In fact, there is a tendency for time-series volatility forecasts to over-predict the specific risk of high-

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volatility stocks, and under-predict the specific risk of low-volatility stocks, as shown in Menchero, Orr, and Wang (2011).

To reduce these biases, we introduce a Bayesian Shrinkage technique. The stocks are segmented into deciles based on their market capitalization. Within each decile, the mean and standard deviation of the specific risk forecasts are computed. The specific risk forecasts are then shrunk toward the mean. The shrinkage intensity for a stock depends on both the distance of its specific risk forecast from the mean and the standard deviation of the specific risk forecasts within the decile.

For more technical details on Bayesian Shrinkage, see Appendix C.

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3 Factor Structure Overview

3.1 Estimation Universe

The coverage universe is the set of all securities for which the model provides risk forecasts. By contrast, the estimation universe is the subset of stocks used to actually estimate the model. Judicious selection of the estimation universe is an important part of building a sound risk model. The estimation universe must be broad enough to accurately represent the investment opportunity set for investors, without being too broad to include illiquid stocks that may introduce spurious return relationships into the model. Furthermore, the estimation universe must be sufficiently stable to ensure that factor exposures are well-behaved across time. Representation, liquidity, and stability, therefore, are the three primary issues to address when selecting a risk model estimation universe.

A well-constructed equity index must address these very issues and therefore serves as an excellent foundation for the estimation universe. The MSCI China A Onshore Index aims to reflect the full breadth of investment opportunities in the China domestic market by targeting the large, mid and small-cap stocks listed on the Shanghai and Shenzhen exchanges. The index construction methodology is similar to that for the MSCI All Country World Investable Markets Index (ACWI IMI). It applies innovative rules designed to achieve index stability, while reflecting the evolving equity markets in a timely manner. Moreover, liquidity screening rules are applied to ensure that only investable stocks with reliable pricing are included for index membership. The China A Onshore Index thus constitutes a natural foundation for the estimation universe of the new Barra China A Equity Models. However, in the history before June 2008, model estimation universe requires some modifications and augmentation to the China A Onshore Index, as described below:

I. The history of the new China A models starts from December 31, 1998. But the China A Onshore IMI Index did not begin until December 2004. Moreover, before June 2008, China A Onshore IMI did not have enough stocks for it to be used as an estimation universe because the size and liquidity of many A shares did not satisfy the index inclusion criteria then. Hence for the period from December 1998 to May 2008, we construct the estimation universe by following a methodology that is similar to the one currently applied by the MSCI ACWI IMI.

II. For the period since June 2008, we directly use the China A Onshore Index as the estimation universe.

An issue that concerns most investors in the China A market is stock suspension. Once a stock is suspended, shares cannot trade until the suspension is lifted or lapses. Stock suspension screening rules are implemented in China A Onshore IMI. For example,

I. MSCI will exclude newly eligible securities (securities that are not current constituents of the Investable Market Indexes) from the investable equity universe if the securities:

   a. Are suspended on the Price Cutoff date, or

   b. Have been suspended for at least 50 days consecutively in the past 12 months

II. Suspended index members breaching the threshold of 50 days will be deleted at the lowest system price

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III. Additions of A shares into the MSCI China A Onshore IMI index will be cancelled if the securities are suspended on the day prior to the effective implementation of the Index Review. The inclusion of the security will be re-evaluated at the next Index Review.

3.2 Market Factor

Similar to the CNE5 Model, the new Barra China A Equity Model includes an explicit Market factor. One significant benefit of the Market factor is the insight and intuition it affords to portfolio managers. Menchero, Orr, and Wang (2011) show that the Market factor portfolio can be interpreted as the capitalization-weighted market portfolio. It separates the market effect and the pure industry effects, thus allowing a more intuitive interpretation of the industry factors.

Without the Market factor, the industry factors represent portfolios that are 100 percent net long the particular industry, with zero net weight in every other industry. With the Market factor, by contrast, the industry factors represent dollar-neutral portfolios that are 100 percent long the industry and 100 percent short the Market factor. As a result, the industry performance is measured net of the market.

The dollar-neutral industry factor portfolios are important for attribution. For instance, suppose that a portfolio manager over-weights an industry that underperforms the overall market, but the industry nonetheless has a positive return. Clearly, overweighting an underperforming industry detracts from performance. If the industry factors are represented by net-long portfolios, however, an attribution analysis would show that overweighting the underperforming industry contributed positively to performance. This non-intuitive result is resolved by introducing the Market factor. Including the Market factor also has benefits in risk attribution, which are fully discussed by Davis and Menchero (2011).

Another benefit of the Market factor is the improvement in risk forecast. Intuitively and empirically, we know that industry indexes tend to become more correlated in times of financial crisis. As shown in Menchero, Orr, and Wang (2011), the Market factor is able to capture these changes in correlation in a more timely manner. The underlying mechanism for this effect is that net-long industry portfolios have common exposure to the Market factor, and when the volatility of the Market factor rises during times of market stress, it explains the increased correlations for those portfolios.

3.3 Industry Factors

Industries are important variables for explaining cross-sectional equity returns. The industry structure is constructed based on the Global Industry Classification Standard (GICS®). In Table 3.1, we report the underlying GICS® codes that map to each of the 32 industry factors of the model. Note that most industry factors of the new Barra China A Equity Model are the same as those of the CNE5 Model. In Table 3.2, we report the average weights of the industries over the sample period July 2004 to March 2018, together with their weights and the largest stocks at the end of the sample period.

Table 3.1: Mapping of the industry factors of the Barra China A Equity Models to GICS® codes

GICS® Sector	Industry Factor Name	GICS® Codes	Change from CNE5

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GICS® Sector	Industry Factor Name	GICS® Codes	Change from CNE5
Energy	Energy	10	
Materials	Commodity Chemicals	15101010	Split from Chemicals
	Non-Commodity Chemicals	15101020, 15101030, 15101040, 15101050	Split from Chemicals
	Construction Materials	151020,	
	Packaging and Paper & Forest Products	151030, 151050	
	Metals and Mining	151040	
Industrials	Aerospace and Defense	201010	
	Building Products	201020	
	Construction & Engineering	201030	
	Electrical Equipment	201040	
	Conglomerates & Distributors	201050, 201070	Merge Conglomerates with Distributors
	Machinery	201060	
	Commercial & Professional Services	2020	
	Airlines	203010, 203020	
	Marine	203030	
	Road & Rail & Transportation Infrastructure	203040, 203050	
Consumer Discretionary	Automobiles & Components	2510	
	Consumer Durables	2520	
	Apparel		
	Consumer Services	2530	
	Media	2540	
	Retailing	2550	
Consumer Staples	Food & Staples Retailing and Household & Personal Products	3010, 3030	
	Beverages & Tobacco	302010, 302030	

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GICS® Sector	Industry Factor Name	GICS® Codes	Change from CNE5
	Food Products	302020	
Health Care	Health Care	35	
Financials	Banks	4010	
	Diversified Financials & Insurance	4020, 4030	
Information Technology & Telecom	Software and Services	4510	
	Hardware and Semiconductors	4520, 4530, 50	
Utilities	Utilities	55	
Real Estate	Real Estate	60	

Table 3.2: Weights of industry factors. The weights are computed using the market capitalizations of the stocks in the estimation universe.

Industry Factor Name	Average Weight (%)	Mar 30, 2018 Weight (%)	Largest Stock & Market Cap in $Bn (Mar 30, 2018)
Energy	9.78	6.59	PETROCHINA CO (196.62)
Commodity Chemicals	4.01	2.09	KANGDE XIN GROUP CO LTD (11.10)
Non-Commodity Chemicals	1.90	1.56	SC TIANQI LITHIUM IND - A (10.69)
Construction Materials	1.18	1.14	ANHUI CONCH CEMENT CO LTD (20.45)
Packaging and Paper & Forest Products	0.83	0.75	SHAN DONG SUN PAPER - A (4.53)
Metals and Mining	7.31	4.77	BAOSHAN IRON & STEEL -A (30.15)
Aerospace and Defense	0.46	0.70	AECC AVIATION POWER CO LTD (10.11)
Building Products	0.47	0.47	BEIJING NEW BLDG MAT -A (7.16)
Construction & Engineering	2.22	3.63	CHN STATE CONSTRUCTION EN (41.29)
Electrical Equipment	2.45	3.13	SZ LUXSHARE PRECISION - A (12.19)
Conglomerates & Distributors	2.55	0.72	XIAMEN C & D INC -A (5.04)
Machinery	4.14	3.51	CRRC CORP LTD (38.86)
Commercial & Professional Services	0.31	0.73	TUS SOUND ENVIRONMENTAL RESOURCES (4.62)
Airlines	0.80	1.05	AIR CHINA LTD (18.76)

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Industry Factor Name	Average Weight (%)	Mar 30, 2018 Weight (%)	Largest Stock & Market Cap in $Bn (Mar 30, 2018)
Marine	0.63	0.33	COSCO SHIPPING HOLDINGS CO LTD (7.15)
Road & Rail & Transportation Infrastructure	3.58	1.94	SH INTL PORT GROUP -A (25.97)
Automobiles & Components	3.52	3.94	SAIC MOTOR CORP LTD (63.16)
Consumer Durables	2.71	3.14	MIDEA GROUP CO LTD (56.99)
Apparel	2.08	1.36	ZHEJIANG SEMIR GARMENT ORD SHS A (4.41)
Consumer Services	0.96	0.79	CHINA INT TRAVEL SER - A (16.42)
Media	0.85	1.07	ORIENTAL PEARL MEDIA CO LTD (6.87)
Retailing	2.79	1.45	SUNING COM CO LTD (20.82)
Food & Staples Retailing and Household & Personal Products	0.62	0.66	YONGHUI SUPERSTORES ORD SHS A (14.97)
Beverages & Tobacco	2.45	3.93	KWEICHOW MOUTAI -A (136.49)
Food Products	2.54	2.80	IN MONGOLIA YILI IND -A (27.52)
Health Care	4.45	6.19	JIANGSU HENGRUI MEDI -A (39.17)
Banks	10.80	15.99	INDUSTRIAL & COMMERCIAL BANK OF CHINA (260.97)
Diversified Financials & Insurance	4.17	7.90	PING AN INSURANCE GRP CO OF CHINA LTD (112.45)
Software and Services	1.17	2.01	IFLYTEK CO LTD (13.43)
Hardware and Semiconductors	5.69	7.27	HZ HIKVISION DIGITAL - A (60.57)
Utilities	5.68	3.11	CHINA YANGTZE POWER -A (56.02)
Real Estate	6.53	5.30	CHINA VANKE CO LTD (51.45)

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3.4 Style Factors

Investment style represents another major source of systematic risk for equity portfolios, especially active portfolios. Style factors are constructed from financially-intuitive stock attributes called descriptors, which serve as effective predictors of equity return covariance.

To facilitate comparison across descriptors and style factors, they are standardized with in the estimation universe. A style factor and its descriptors are always standardized on the same basis. Specifically, if $x_{n,k}^{Raw}$ is the raw value of stock n for descriptor (or factor) k, then the standardized descriptor (or factor) value is given by:

$$x_{n,k} = \frac{x_{n,k}^{Raw} - \mu_k}{\sigma_k}, \tag{3.1}$$

where σ_k is the equal-weighted standard deviation of $x_{n,k}^{Raw}$ within the entire estimation universe, and μ_k is the cap-weighted mean of $x_{n,k}^{Raw}$ within the entire estimation universe. As a result of standardizing using the cap-weighted mean, a portfolio well-diversified across China A, such as the China A Onshore Index, has approximately zero exposure to all style factors.

Below, we provide a brief qualitative description of each of the 16 style factors in the new Barra China A Long-Term Model:

- **Beta** – Captures market risk that cannot be explained by the Market factor. To better understand this factor, consider a fully invested long-only portfolio strongly tilted toward high-beta stocks. Intuitively, this portfolio has greater market risk than a portfolio with beta equal to one. This additional market risk is captured through a positive exposure to the Beta factor. The time-series correlation between the Market factor and the Beta factor is typically very high, and therefore these two sources of risk reinforce each other in this example. If, by contrast, the portfolio is invested in low-beta stocks, then the risk from the Beta and the Market factors is partially cancelled, as expected.
- **Book-to-Price** – is considered by some to be an indicator of value. It specifies how inexpensively a company is currently traded by using its book value as a yardstick.
- **Dividend Yield** – Differentiates stocks based on their trailing 12-month and predicted dividend-to-price ratios.
- **Earnings Quality** – Captures return differences due to the accrual components of earnings.
- **Earnings Variability** – Captures the variability in earnings and cash flows using both historical measures and analyst predictions.
- **Earnings Yield** – Differentiates stocks based on the earnings of the companies relative to their prices. Earnings Yield is considered by many investors to be a strong value signal. The most important descriptor in this factor is the analyst-predicted earnings-to-price.
- **Growth** – Differentiates stocks based on their prospects for sales or earnings growth. The most important descriptor in this factor is the analyst-predicted long-term earnings growth. It also includes the sales and earnings growth over the trailing five years.
- **Investment Quality** – Combines assets, capital expenditure, and net issuance growth.

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- **Leverage** – Captures return differences between high-leverage and low-leverage stocks. The descriptors within this factor include market leverage, book leverage, and debt-to-assets ratio.
- **Liquidity** – Captures return differences due to relative trading activity, measured by the fractions of total shares outstanding that are traded over selected trailing windows.
- **Long-Term Reversal** – Differentiates stocks based on long-term (four years lagged by thirteen months) relative performance. It is orthogonalized to the Momentum factor.
- **Mid Capitalization** – Captures non-linearity in the payoff to the Size factor across the market-cap spectrum. This factor is based on a single raw descriptor: the cube of the Size exposure. However, because this raw descriptor is highly collinear with the Size factor, it is orthogonalized to the Size factor. This procedure does not affect the fit of the model, but does mitigate the confounding effects of collinearity, thus preserving an intuitive meaning for the Size factor. As described by Menchero (2010), this factor roughly captures the risk of a "barbell portfolio" that is long mid-cap stocks and short small and large-cap stocks.
- **Momentum** – Differentiates stocks based on their recent 12-month performance. In computing this factor, we exclude the returns in the most recent month to avoid the effects of short-term reversal.
- **Profitability** – Combines profitability measures that characterize efficiency of a firm's operations and total activities.
- **Residual Volatility** – Consists of three descriptors: the volatility of daily excess returns, the volatility of daily residual returns, and the cumulative range of the stock excess return over the last 12 months. As these descriptors tend to be highly collinear with the Beta factor and also with the Size factors to a lesser but still significant degree, this factor is orthogonalized to these two factors.
- **Size** – Represents another strong source of equity return covariance and captures return differences between large and small-cap stocks. We measure Size by the log of market capitalization.

For information about factor and descriptor analysis, see Appendix D. For descriptor definitions, see Appendix E.

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4 Model Characteristics and Properties

4.1 Factor Statistics

One requirement for a high-quality factor structure is that the factor returns be statistically significant. This helps prevent weak or noisy factors from finding their way into the model. We measure statistical significance by the t-statistic of the cross-sectional regressions. Assuming normality, absolute t-statistics greater than two are considered significant at the 95-percent confidence level. In other words, even if the factor had no explanatory power (that is, the factor is pure noise), there is a chance that we would still observe a t-statistic above two about five percent of the time.

In Table 4.1, we report summary statistics for the style factor returns from the Barra China A Long-Term Model for the sample period July 2004 to March 2018. Monthly regression results are used to compute the statistics. We also report the Factor Stability Coefficient and Variance Inflation Factor (VIF) for the factors.

The Factor Stability Coefficient is computed as the cross-sectional correlation of factor exposures from one month to the next. The Variance Inflation Factor measures the extent a factor can be explained by the other factors in the model, a property often referred to as multi-collinearity. Excessive multi-collinearity can lead to high VIF and increased estimation error in the factor returns and non-intuitive correlations among factors. As shown in the table, all the style factors are below this level during the sample period. In Table 4.2, we report summary statistics for industry factor returns.

Table 4.1: Style factor summary statistics

Style Factor	Average |t-stat|	Percent |t|>2	Annual Return (%)	Annual Volatility (%)	IR	Corr. with Market	Factor Stability Coeff.	VIF
Size	4.02	70.2	-7.59	6.28	-1.21	0.01	0.99	6.41
Momentum	2.96	57.3	1.80	4.88	0.37	-0.12	0.91	2.73
Residual Volatility	2.75	57.3	-0.89	4.78	-0.19	0.29	0.95	2.70
Beta	3.02	57.9	2.48	4.66	0.53	0.54	0.97	2.48
Liquidity	2.61	53.2	-6.78	3.49	-1.94	0.33	0.97	4.62
Earnings Yield	1.75	29.2	1.94	3.20	0.61	0.26	0.97	4.62
Book-to-Price Ratio	1.69	34.5	1.90	3.20	0.59	0.14	0.97	3.31
Mid-Capitalization	2.14	44.4	-3.24	2.66	-1.22	0.09	0.97	1.39
Long-Term Reversal	1.43	24.6	0.60	2.29	0.26	0.19	0.93	2.06
Earnings Variability	1.67	35.0	0.11	2.27	0.05	0.16	0.97	1.46
Profitability	1.28	17.0	1.15	2.23	0.52	-0.23	0.99	3.15

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Style Factor	Average |t-stat|	Percent |t|>2	Annual Return (%)	Annual Volatility (%)	IR	Corr. with Market	Factor Stability Coeff.	VIF
Leverage	1.24	20.5	-0.00	1.92	-0.00	-0.02	0.99	2.09
Investment Quality	1.20	19.6	-0.06	1.84	-0.03	0.06	0.98	1.44
Dividend Yield	0.94	9.9	0.60	1.40	0.43	0.04	0.96	2.86
Growth	1.03	12.9	1.13	1.35	0.84	-0.23	0.96	1.66
Earnings Quality	1.14	15.8	0.43	1.25	0.35	-0.02	0.93	1.24

Table 4.2: Industry factor summary statistics

Industry Factor	Average |t-stat|	Percent |t-stat|>2	Annual Return (%)	Annual Volatility (%)	IR	Corr. with Market
Energy	2.2	45.6	-2.8	13.5	-0.2	0.1
Commodity Chemicals	1.4	25.4	-3.6	9.0	-0.4	-0.1
Non-Commodity Chemicals	1.7	32.5	0.5	11.3	0.0	-0.2
Construction Materials	1.7	30.8	3.5	17.2	0.2	0.1
Packaging and Paper & Forest Products	1.1	15.4	-6.6	9.5	-0.7	-0.2
Metals and Mining	2.3	46.2	2.8	11.2	0.3	0.2
Aerospace and Defense	1.8	30.8	14.8	24.7	0.6	0.0
Building Products	1.0	10.7	-2.4	13.0	-0.2	-0.2
Construction and Engineering	1.6	27.2	-0.1	11.6	0.0	0.0
Electrical Equipment	1.5	26.6	3.5	11.4	0.3	-0.1
Conglomerates and Distributors	1.2	17.2	0.3	9.4	0.0	0.1
Machinery	1.8	39.1	2.0	9.3	0.2	0.0
Commercial and Professional Services	1.1	13.0	-0.1	17.9	0.0	-0.2
Airlines	1.7	38.5	0.4	21.5	0.0	0.1
Marine	1.3	18.3	-2.5	20.9	-0.1	0.1
Road & Rail & Transportation Infrastructure	1.7	29.6	-7.0	11.5	-0.6	-0.2

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Industry Factor	Average |t-stat|	Percent |t-stat|>2	Annual Return (%)	Annual Volatility (%)	IR	Corr. with Market
Automobiles and Components	1.8	37.9	-0.0	10.9	-0.0	0.0
Consumer Durables	1.6	29.6	3.6	12.0	0.3	-0.2
Apparel	1.3	18.3	-2.8	9.3	-0.3	0.0
Consumer Services	1.2	16.6	-2.3	12.2	-0.2	-0.2
Media	1.4	24.3	4.0	16.5	0.2	-0.1
Retailing	1.6	30.2	-1.7	10.6	-0.2	0.0
Food & Staples Retailing and Household & Personal Prod	1.1	16.0	-2.9	12.6	-0.2	-0.1
Beverages and Tobacco	1.8	38.5	5.9	14.5	0.4	-0.1
Food Products	1.6	32.5	0.4	10.3	0.0	-0.3
Health Care	2.1	41.4	5.4	11.7	0.5	-0.2
Banks	1.9	38.5	2.1	16.7	0.1	-0.2
Diversified Financials & Insurance	2.5	49.1	14.4	25.7	0.6	0.3
Software and Services	1.8	34.9	9.7	15.4	0.6	0.0
Hardware and Semiconductors	2.3	45.0	1.9	11.2	0.2	-0.2
Utilities	1.8	35.5	-7.6	10.8	-0.7	-0.1
Real Estate	2.6	50.9	3.5	14.6	0.2	0.0

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4.2 Explanatory Power

The explanatory power of the factors is a key measure of model quality. However, it can be significantly impacted by the regression weighting scheme, the estimation universe, and the time period under consideration. Caution is required when comparing the explantory power across different models. Nevertheless, if each of these variables is carefully controlled, a meaningful comparison between models is possible.

There are various ways of defining the explanatory power of the factors. We use the leave-one-out Cross-Validated R^2 ($CV\ R^2$). In principle, the leave-one-out cross-validation process is conducted as follows[4]:

1. For each period, for each stock in the estimation universe, we generate a *custom* set of factor returns by keeping this stock out of the estimation universe, and then performing the regression with the rest *N* - 1 stocks. Here *N* denotes the total number of stocks in the estimation universe. Because a stock cannot exert any influence on its own *custom* set of factor returns, this set of factor returns can be considered as being estimated *out-of-sample* for this stock. We can then calculate the residual return for this stock using its own *custom* set of factor returns. Finally, we can calculate the $CV\ R^2$ for this period:

$$CV\ R^2 = 1 - \frac{\sum_n w_n u_n^2}{\sum_n w_n r_n^2} \tag{4.1}$$

   where r_n is the excess return of stock n, u_n is the residual return of this stock calculated as described above, and w_n is the regression weight of this stock.

2. Repeat the above process for the entire model history. We thus obtain a time series of $CV\ R^2$ for the model.

Another measure of explanatory power is $Studentized\ R^2$. The model that produces a higher Studentized R^2 is considered to have a smaller over-fitting or under-fitting risk. The Studentized R^2 is calculated as follows:

1. For each period, let X be the matrix of factor exposure, then $H = X(X'X)^{-1}X'$ is the orthogonal projection on the column space of X. We calculate the Studentized R^2 for this period as:

$$Studentized\ R^2 = 1 - \frac{\sum_n w_n \frac{\varepsilon_n^2}{1-H_{nn}}}{\sum_n w_n r_n^2}, \tag{4.2}$$

   where r_n is the excess return of stock n, ε_n is the residual return of this stock from the cross-sectional regression, w_n is the regression weight of this stock, H_{nn} is n-th diagonal element of the matrix H.

2. Repeat the above process for the entire model history. We thus obtain a time series of $Studentized\ R^2$ for the model.

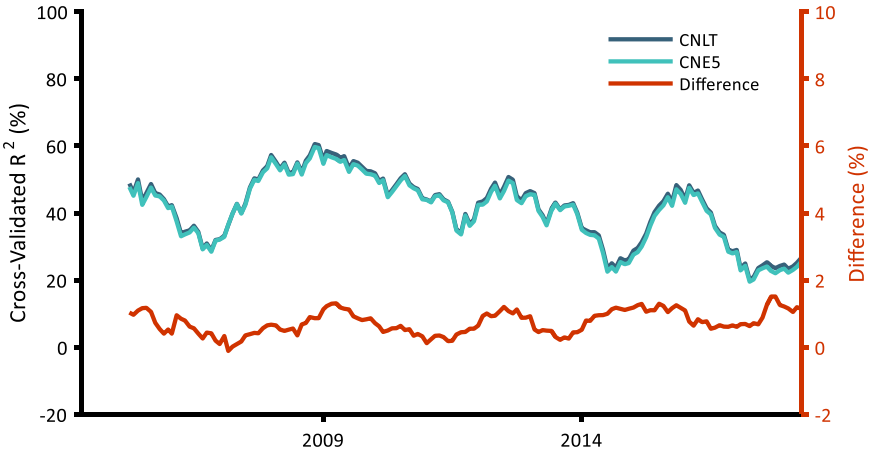
- We conduct a comparison of the $CV\ R^2$ and the $Studentized\ R^2$, respectively, between the new Barra China A Long-Term Model, and the CNE5 model, using monthly regressions for the sample period July 2004 to March 2018. The estimation universe of the Barra China A Long-Term Model is used as the estimation universe for this comparison. The same regression weighting scheme as in the Barra China A Long-Term Model is used. The only input that is varied is the factor exposure matrix.

[4] In practice, this measure can be calculated analytically. For more discussions about $CV\ R^2$, see Wasserman, L. (2004).

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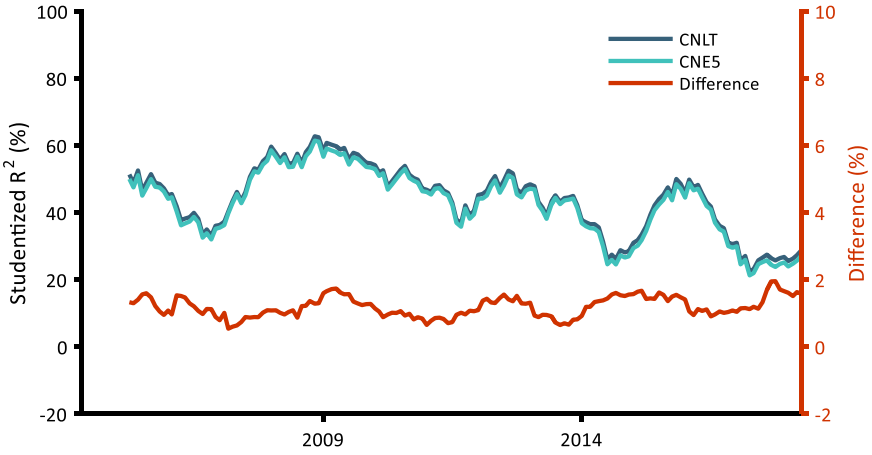
In Figure 4.1, we plot the rolling 12-month $CV\ R^2$ for the two models. We see that through most of the sample period, the Barra China A Long-Term Model provides a higher $CV\ R^2$ than the CNE5 Model. The difference between the $CV\ R^2$'s for the two models varies largely over time. On average, the Barra China A Long-Term Model has an advantage of about 76 bps over the CNE5 Model.

Figure 4.1: Rolling 12-month Cross-Validated R^2 for the Barra China A Long-Term Model and the CNE5 Model



A similar comparison of the rolling 12-month $Studentized\ R^2$ is shown in Figure 4.2. The average difference of $Studentized\ R^2$ between the Barra China A Long-Term Model and the CNE5 model is 120 bps.

Figure 4.2: Rolling 12-month Studentized R^2 for the Barra China A Long-Term Model and the CNE5 Model



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4.3 Cross-Sectional Dispersion

The cross-sectional dispersion of stock returns can be measured in two ways:

- by cross-sectional volatility (CSV), which measures the dispersion relative to the mean return, or
- by root mean square (RMS) return, which measures the dispersion relative to zero return

The main difference between the two is that the Market factor does not make contribution to the CSV, but does to the RMS return. As discussed by Menchero and Morozov (2011), the RMS return can be decomposed and attributed to individual factors or groups of factors (see Appendix F).

In Figure 4.3, we plot the rolling 12-month contributions to the monthly RMS return by factors and specific return for the Barra China A Long-Term Model. While the stock-specific contribution dominates during the financial crisis, factors are more important during the most recent period.

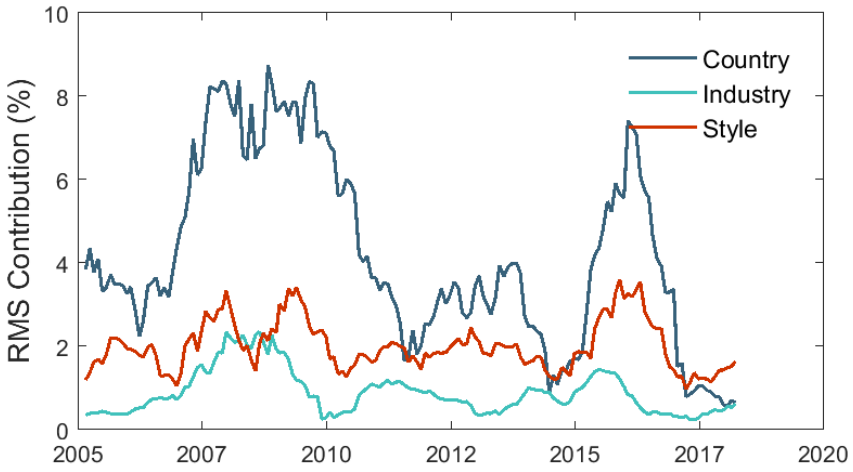
Figure 4.3: Rolling 12-month contributions to the monthly root mean square (RMS) return by factor and specific return for the Barra China A Long-Term Model



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In Figure 4.3, we further decompose the factor RMS return into the Market, industry, and style components. As shown, the Market factor is the largest contributor to the factor RMS return through most of the sample period. The style factors contribute more than the industries, except for a short period during the financial crisis.

Figure 4.3: Rolling 12-month contributions to the monthly root mean square (RMS) return by factor types for the Barra China A Long-Term Model



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5 Risk Forecasting Accuracy

5.1 Portfolio Risk Forecasting Accuracy

In this section, we compare the risk forecasting accuracy of the new Barra China A Long-Term Model and its predecessor, the CNE5 Model. Our methodology for evaluating and comparing the accuracy of forecasts is based on the Q-statistic and the bias statistic. We use the Q-statistic to quantify the differences between models, and the bias statistic to build intuition about the periods when a model under-forecasts or over-forecasts risk. For a more technical discussion of measures of bias, see Appendix G.

The bias statistic is an out-of-sample measure that represents the ratio of realized risk to predicted risk. The bias statistic for perfect risk forecasts is one. By plotting the mean rolling-window bias statistic across time for a collection of portfolios, we can visualize the magnitude of the average biases and judge if they are persistent or regime-dependent.

One shortcoming of the bias statistic is that over a long period, we may have sub-periods of over-forecasting and under-forecasting, yet obtain a bias statistic close to one over the entire period. Forecasting errors can cancel out over the long term, even though the accuracy may be poor over sub-periods. For this reason, we focus on the Q-statistic, which provides a measure of the forecast error and grows with the error size. The Q-statistic is not prone to the error cancellation and is minimized by having the exact forecast for every portfolio in every time period. This gives us a tool to measure the improvements between models on the same set of portfolios. The more accurate model will have a lower average Q-statistic.

A variety of test portfolios are constructed to compare the risk forecasting accuracy of the Barra China A Long-Term Model and the CNE5 Model. The estimation universe of the Barra China A Long-Term Model is used as the investment universe from which the test portfolios are constructed. The portfolios are updated monthly and bias statistics and Q-statistics are computed from monthly returns and monthly risk forecasts. All the tests are conducted for the sample period July 2004 to March 2018.

Table 5.1 summarizes the risk forecasting accuracy test results for the portfolios by the two models. The results for each category of portfolios are averaged over all the portfolios in the category, and over the entire test period. For all categories, Barra China A Long-Term Model has a clear advantage over the CNE5 Model, judging by both the bias statistics and Q-statistics.

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Table 5.1: Portfolio bias statistics and Q-statistics

Portfolio	Bias				Q				Vol (%)
	CNLTS (Responsive)	CNLTL (Stable)	CNE5S	CNE5L	CNLTS (Responsive)	CNLTL (Stable)	CNE5S	CNE5L	
Random	1.06	1.02	1.15	1.09	2.4675	2.5162	2.5119	2.5414	31.90
Random Active	1.01	1.01	1.06	1.04	2.4767	2.5168	2.4951	2.5366	10.71
CNLT Industry-Tilt	1.05	1.02	1.13	1.10	2.4792	2.5187	2.5210	2.5531	35.81
CNE5 Industry-Tilt	1.05	1.03	1.13	1.10	2.4849	2.5297	2.5236	2.5637	35.99
CNLT Industry-Tilt Active	1.08	1.08	1.16	1.14	2.5908	2.6391	2.6504	2.6962	20.33
CNE5 Industry-Tilt Active	1.08	1.08	1.15	1.13	2.5965	2.6356	2.6473	2.6917	20.42
CNLT Style-Tilt	1.07	1.03	1.16	1.11	2.5001	2.5279	2.5445	2.5643	31.99
CNE5 Style-Tilt	1.08	1.03	1.16	1.11	2.5103	2.5380	2.5557	2.5696	31.84
CNLT Style-Tilt Active	1.06	1.07	1.17	1.16	2.5448	2.6140	2.6267	2.6959	12.33
CNE5 Style-Tilt Active	1.08	1.08	1.15	1.14	2.6211	2.6527	2.6713	2.7235	12.37
CNLT Style Spread	1.08	1.06	1.31	1.29	2.3491	2.3973	2.5652	2.5818	13.28
CNE5 Style Spread	1.11	1.10	1.16	1.14	2.4246	2.4629	2.4986	2.5182	14.39
CNLTS Style Optimized	1.10	1.06	1.14	1.10	2.4763	2.4822	2.5039	2.4993	4.99
CNLTL Style Optimized	1.09	1.06	1.13	1.11	2.4706	2.4785	2.4976	2.4948	4.97
CNE5S Style Optimized	1.13	1.09	1.16	1.12	2.5113	2.5036	2.5376	2.5262	5.72
CNE5L Style Optimized	1.12	1.09	1.16	1.13	2.4877	2.4878	2.5160	2.5105	5.70

The results for each row are averaged over all the portfolios in the category.
(CNLTS: Barra China A Long-Term Model, Responsive Variant)

(CNLTL: Barra China A Long-Term Model, Stable Variant)

To illustrate how the forecasting accuracy of the two models change over time, in the following figures, we plot the average rolling 12-month bias statistics for selected portfolios in Table 5.1.

In Figure 5.1, we plot the average rolling 12-month bias statistics for 100 random long portfolios. Each of these portfolios is constructed by going long 100 stocks randomly selected from the full investment universe and

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weighted by their market capitalizations. To reduce turnover, the stocks used to construct each portfolio are fixed unless a stock drops out of the investment universe, in which case the stock is replaced by another randomly selected stock.

In Figure 5.2, we plot the average rolling 12-month bias statistics for 100 random active portfolios. These portfolios are constructed by going long the 100 random long portfolios used in the test above, and the benchmark is the cap-weighted investment universe.

Figure 5.1: Average rolling 12-month bias statistics for random long portfolios

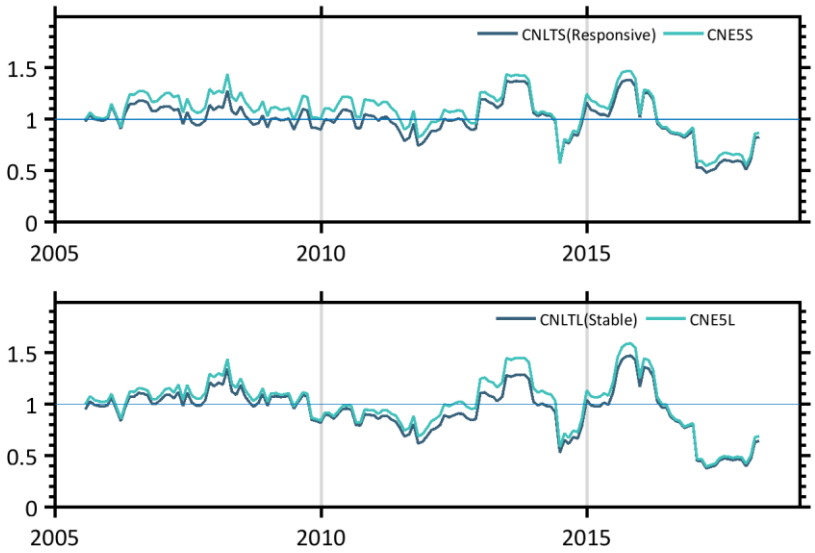
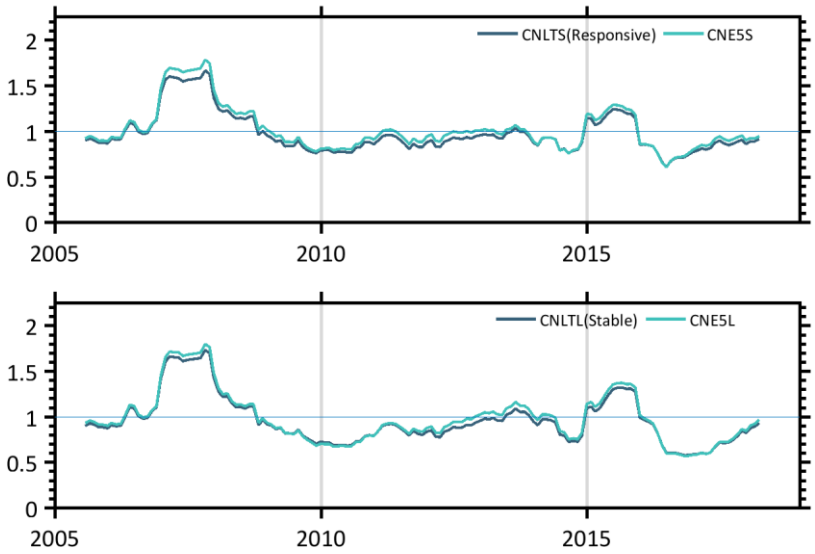


Figure 5.2: Average rolling 12-month bias statistics for random active portfolios



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In Figure 5.3, we plot the average rolling 12-month bias statistics for the industry long portfolios. Each of these portfolios holds the stocks from a separate industry. The stocks are weighted by their market capitalizations.

In Figure 5.4, we plot the average rolling 12-month bias statistics for the industry active portfolios. These portfolios are constructed by going long the industry long portfolios used in the test above, and going short the cap-weighted investment universe.

Figure 5.3: Average rolling 12-month bias statistics for industry long portfolios

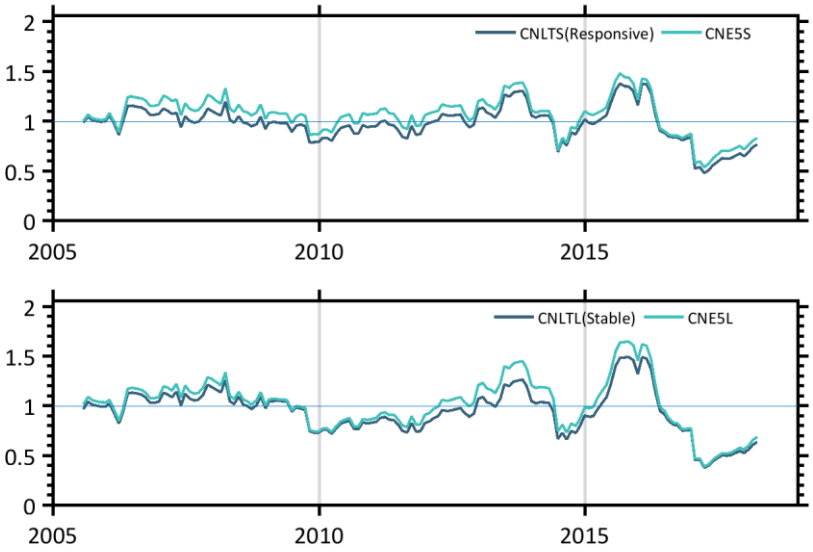
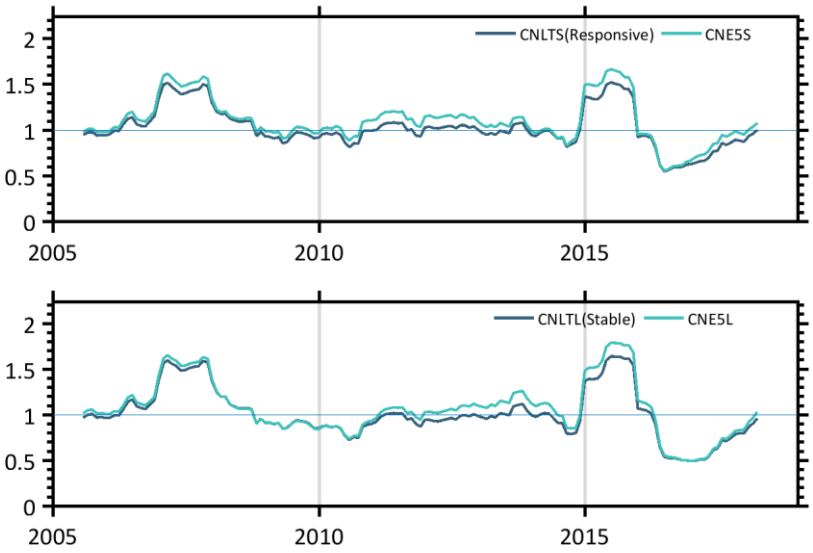


Figure 5.4: Average rolling 12-month bias statistics for industry active portfolios



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In Figure 5.5, we plot the average rolling 12-month bias statistics for the CNLT style-tilt portfolios. For each style factor of the Barra China A Long-Term Model, two cap-weighted portfolios are constructed from the top and bottom quintiles of this style respectively, so there are 36 such portfolios in total in this category.

In Figure 5.6, we plot the average rolling 12-month bias statistics for the CNLT style-tilt active portfolios. These portfolios are constructed by going long the CNLT style-tilt portfolios used in the test above, and going short the cap-weighted investment universe.

Figure 5.5: Average rolling 12-month bias statistics for CNLT-style-tilt portfolios

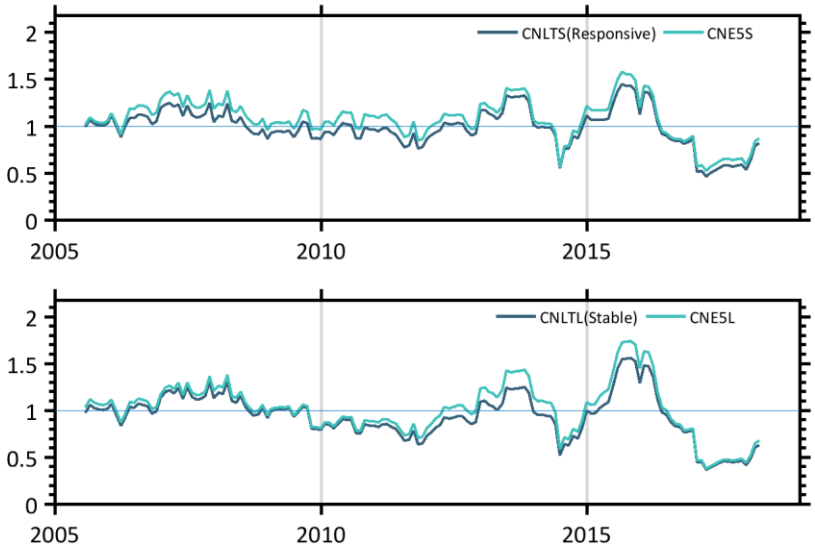
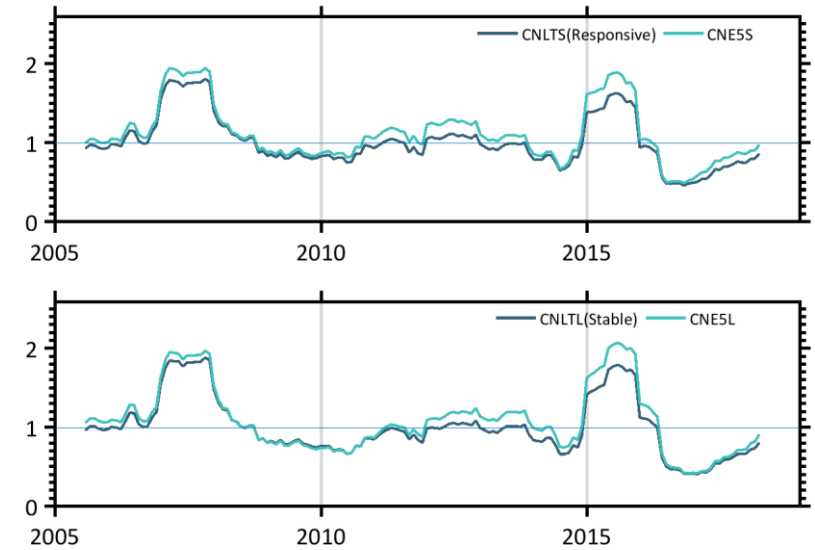


Figure 5.6: Average rolling 12-month bias statistics for CNLT style-tilt active portfolios



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Table 5.2 shows the factor-by-factor risk forecasting accuracy test results for CNLT style-tilt active portfolios. We see that both variants of the new China A Long-Term model provide more risk forecasting accuracy for most portfolios.

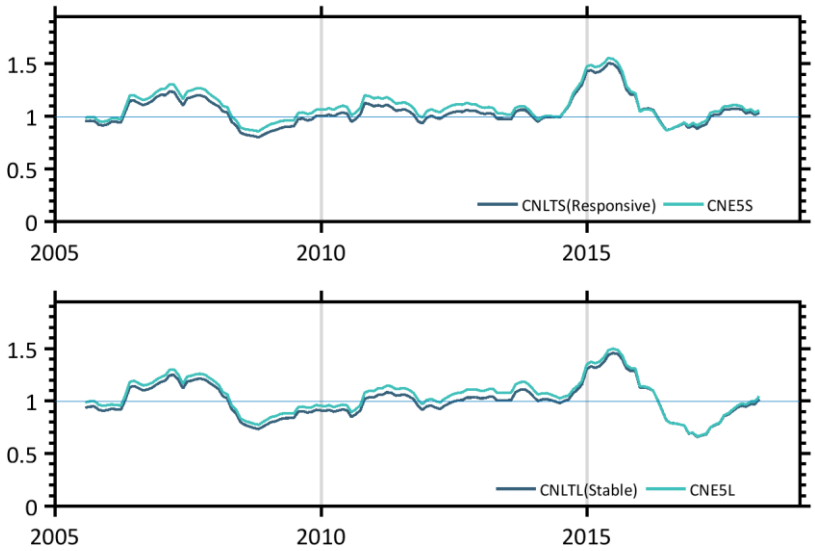
Table 5.2: Bias statistics and Q-statistics of the style-tilt active portfolios

Portfolio	Bias				Q				Vol (%)
	CNLTS (Responsive)	CNLTL (Stable)	CNE5S	CNE5L	CNLTS (Responsive)	CNLTL (Stable)	CNE5S	CNE5L	
Beta	1.07	1.05	1.19	1.15	2.6919	2.7264	2.7808	2.8163	13.24
Book-to-price	1.10	1.12	1.25	1.25	2.5404	2.5851	2.6767	2.7126	13.79
Dividend Yield	1.06	1.07	1.17	1.16	2.4364	2.4987	2.4978	2.5539	11.23
Earnings Quality	0.97	0.95	1.06	1.04	2.4992	2.5639	2.5369	2.6366	7.89
Earnings Variability	1.14	1.14	1.28	1.26	2.4398	2.5443	2.5770	2.6529	11.87
Earnings Yield	1.09	1.10	1.21	1.21	2.5633	2.6460	2.6375	2.7410	13.32
Growth	1.06	1.07	1.16	1.17	2.5281	2.6069	2.6226	2.7013	11.72
Investment Quality	1.06	1.07	1.20	1.19	2.4781	2.5331	2.6145	2.6437	11.23
Leverage	1.11	1.12	1.21	1.21	2.7932	2.8628	2.8609	2.9713	11.68
Liquidity	1.06	1.08	1.16	1.15	2.6402	2.6997	2.6847	2.7703	13.56
Long-Term Reversal	1.14	1.15	1.29	1.30	2.7048	2.7736	2.8657	2.9186	12.92
Mid Capitalization	1.12	1.15	1.20	1.21	2.5482	2.6802	2.6286	2.7579	15.63
Momentum	1.03	1.03	1.13	1.11	2.4614	2.5308	2.5148	2.5349	13.76
Profitability	0.96	0.94	1.03	1.00	2.4666	2.4961	2.4681	2.5089	9.83
Residual Volatility	0.90	0.90	0.98	0.97	2.3868	2.4090	2.3804	2.4176	10.68
Size	1.14	1.17	1.27	1.28	2.5765	2.7036	2.7324	2.8180	15.03

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In Figure 5.7, we plot the average rolling 12-month bias statistics for the CNLTS (Responsive) style-optimized portfolios. A style factor of the Barra China A Long-Term Model as the "alpha signal", and the Responsive variant of the Barra China A Long-Term Model as the risk model, we can analytically solve for the minimum-volatility portfolio subject to the unit-alpha constraint (this portfolio is also called the characteristic portfolio for this alpha signal on this stock universe). The style-optimized portfolios are recalculated on each month end. For each style factor, we use 20 random portfolios in Figure 5.1 as the investment universe. There are 320 such style-optimized portfolios to be constructed for the 16 style factors of the Barra China A Long-Term Model.

Figure 5.7: Average rolling 12-month bias statistics for CNLTS (Responsive) style-optimized portfolios



5.2 Beta Forecasting Accuracy

In this section, we compare the forecasting accuracy of the predicted betas computed using the new Barra China A Long-Term Model and the CNE5 Model. Specifically, following the methodology discussed by Menchero, Nagy, and Singh (2014), for each model, we first compute the regression-weighted cross-sectional variance of the out-of-sample residual returns of the stocks. Here the out-of-sample residual return of stock n for month t is given by:

$$u_{n,t} = r_{n,t} - \beta_{n,t-1} r_{m,t} \,, \tag{5.1}$$

where $r_{n,t}$ and $r_{m,t}$ are the excess returns of the stock and the market for the month, respectively, and $\beta_{n,t-1}$ is the predicted beta of the stock computed using the model at the previous month end. The out-of-sample residual R-Squared is the regression-weighted cross-sectional variance of the residual returns divided by that of the stock excess returns:

$$RR2_t = \frac{\sum_n w_n u_{n,t}^2}{\sum_n w_n r_{n,t}^2}, \tag{5.2}$$

where w_n is the regression weight for stock n. Over a long model history, we obtain a time series of monthly out-of-sample residual R-Squared for each model.

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In Figure 5.8, we report the cumulative difference of the monthly out-of-sample residual R-Squared for the Responsive (S) variants between the Barra China A Long-Term Model and the CNE5 Model, divided by the total number of month. A similar chart for the Stable (S) variants is shown in Figure 5.9. We see that in most of the test period, the Barra China A Long-Term Model gives a lower out-of-sample residual R-Squared than the CNE5 Model, for both the Responsive and Stable variants. As results, the new China A Long-Term model provides more accurate forecast betas.

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Figure 5.8: The cumulative difference of out-of-sample residual R-Squared (%) for the Responsive (S) variants, divided by the total number of months

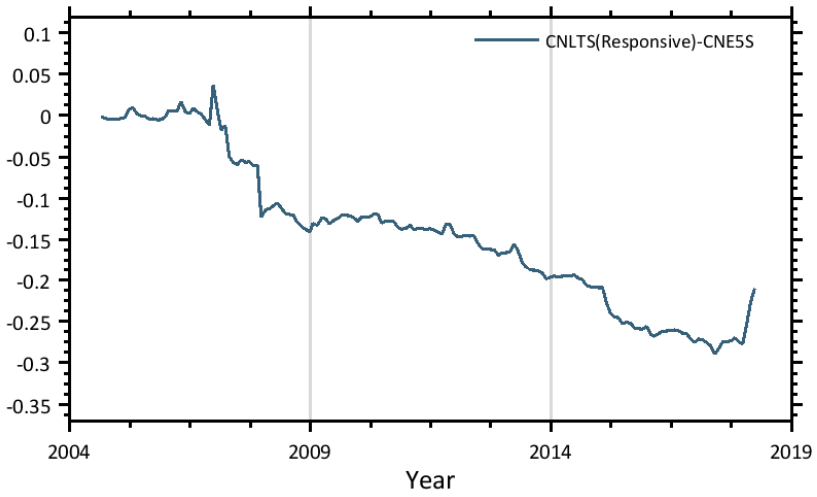
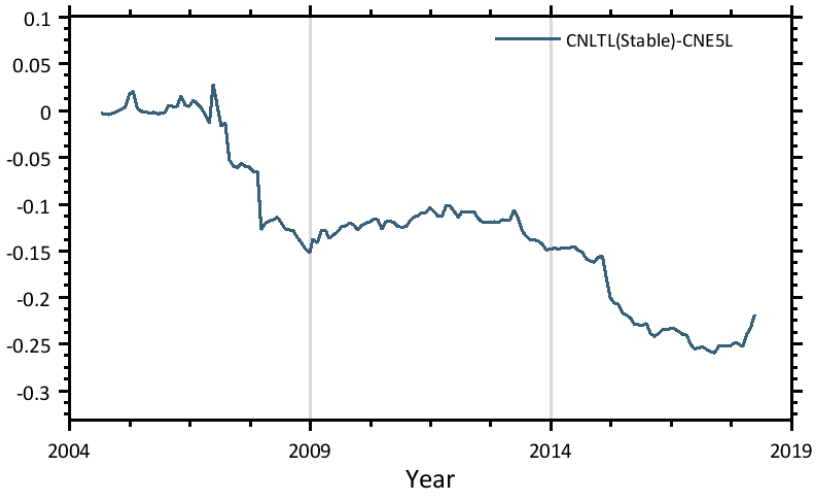


Figure 5.9: The cumulative difference of out-of-sample residual R-Squared (%) for the Stable (L) variants, divided by the total number of months.



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6 Portfolio Performance

In this section, we compare the performance of the portfolios constructed using the new Barra China A Long-Term Model with those using the CNE5 model.

6.1 Analytical Minimum-Volatility Portfolio

In Figure 6.1, we plot the average rolling 12-month volatility of the analytical minimum-volatility fully-invested portfolios. The investment universes of these portfolios correspond to the random portfolios in Figure 5.1. As such, there are 20 portfolios, each of which has 100 stocks. These portfolios are generated using the Responsive (S) variant of the new China A Long-Term Model and CNE5, respectively.

Figure 6.1: Rolling 12-month realized volatility of analytical minimum-volatility portfolio

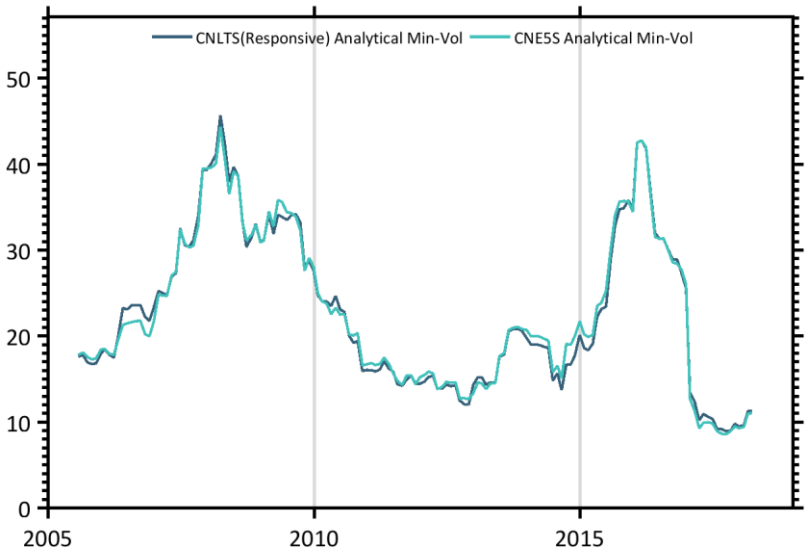


Table 6.1: Summary statistics of analytical minimum-volatility portfolios

Portfolio	Bias				Q				Vol (%)
	CNLTS (Responsive)	CNLTL (Stable)	CNE5S	CNE5L	CNLTS (Responsive)	CNLTL (Stable)	CNE5S	CNE5L	
CNLTS(Responsive) Analytical Min-Vol	1.25	1.16	1.31	1.24	2.5688	2.5507	2.6359	2.6132	26.32
CNE5S Analytical Min-Vol	1.21	1.12	1.41	1.33	2.5922	2.5579	2.7922	2.7387	26.44

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6.2 Backtest Long-only Minimum-Volatility Portfolios

Two types of long-only minimum-volatility portfolios are often constructed by investment managers in practice: one with a constraint on the predicted beta for the portfolio and one without such a constraint. We construct both types of portfolios using the two models in this test. The estimation universe of the Barra China A Long-Term Model is used as the investment universe. For the predicted-beta-constrained minimum-volatility portfolios, a lower bound of 0.5 is set on the predicted beta for the portfolio. The portfolios are rebalanced monthly with an 8% turnover limit. The test period is July 2004 through March 2018.

In Table 6.2, we report the realized volatility, bias statistics, and Q-statistics for the portfolios constructed using the Responsive (S) variants of the two models. We see that a risk reduction of 160 bps is achieved for both the predicted-beta-constrained and unconstrained minimum-volatility portfolios by using the new model.

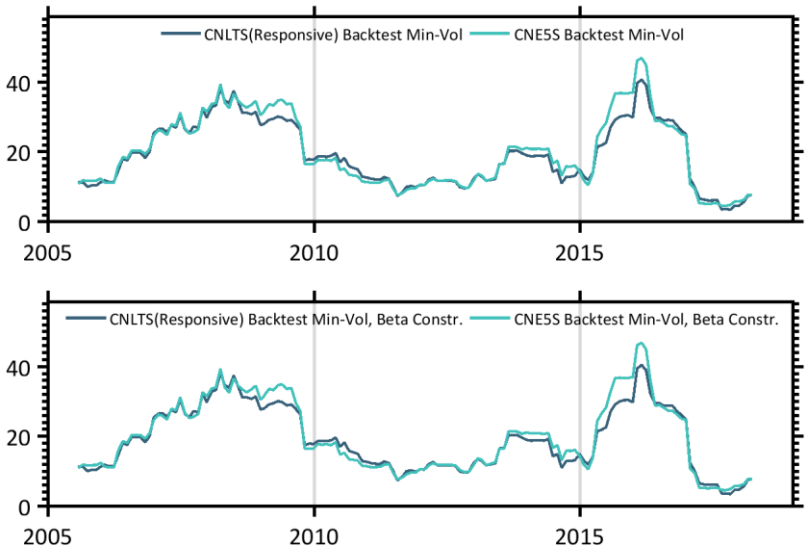
Table 6.2: Summary statistics for long-only minimum-volatility portfolios

Portfolio	Bias		Q		Vol (%)
	CNLTS (Responsive)	CNE5S	CNLTS (Responsive)	CNE5S	
CNLTS (Responsive) Min-Vol	1.12	1.17	2.5181	2.5028	22.58
CNLTS (Responsive) Min-Vol, Beta Constrained	1.12	1.17	2.5565	2.5353	22.56
CNE5S Min-Vol	1.15	1.30	2.7115	2.8268	24.16
CNE5S Min-Vol, Beta Constrained	1.15	1.30	2.7114	2.8267	24.16

It is also interesting to study how the realized volatilities of the portfolios change over time. In Figure 6.1, we plot the rolling 12-month volatilities for the four portfolios listed in Table 6.2: the two predicted-beta-unconstrained minimum-volatility portfolios in the top subplot, and the two predicted-beta-constrained minimum-volatility portfolios in the bottom subplot. We see that the rolling 12-month volatilities for all the four portfolios follow a similar pattern over time, and vary within a large range of about 4-40%.

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Figure 6.1: Rolling 12-month volatilities for long-only minimum-volatility portfolios



6.3 Backtest Long-only Benchmark-Tracking

In this set of tests, we use the market capitalization-weighted estimation universe of the China A Long-Term model as the investment universe and the benchmark portfolio, and construct long-only portfolios with 25 and 40 names, respectively, to track the benchmark. The portfolios are rebalanced monthly with an 8% turnover limit. The test period is July 2004 through March 2018.

In Table 6.3, we summarize the tracking error, bias statistics, and Q-statistics for the portfolios constructed using the Responsive (S) variants of the two models, respectively.

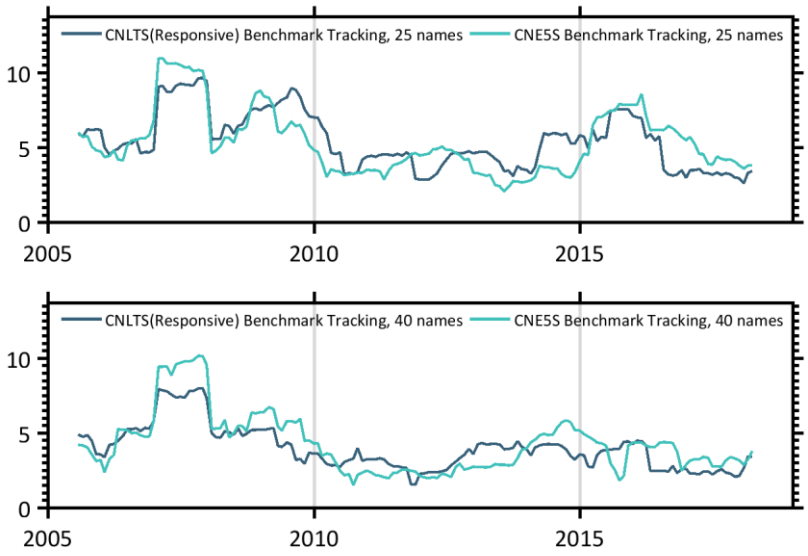
Table 6.3: Summary statistics for long-only benchmark-tracking portfolios

Portfolio	Bias		Q		Tracking Error (%)
	CNLTS (Responsive)	CNE5S	CNLTS (Responsive)	CNE5S	
CNLTS (Responsive) Benchmark-Tracking, 25 names	1.24	1.21	2.5334	2.4994	1.24
CNLTS (Responsive) Benchmark-Tracking, 40 names	1.29	1.25	2.6530	2.6253	1.29
CNE5S Benchmark-Tracking, 25 names	1.17	1.24	2.3557	2.4161	1.17
CNE5S Benchmark-Tracking, 40 names	1.28	1.35	2.6735	2.7409	1.28

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In Figure 6.3, we plot the rolling 12-month tracking error for the four portfolios listed in Table 6.3: the two portfolios with 25 names in the top subplot, and the two with 40 names in the bottom subplot. We see that the rolling 12-month tracking errors for all the four portfolios vary largely over time. However, the two lines plotted in the top subplot have higher tracking errors than the two lines in the bottom subplot. This is understandable because it is easier to track the benchmark when there are more stocks to work with.

Figure 6.3: Rolling 12-month tracking errors of benchmark-tracking portfolios



6.4 Backtest Active Strategy

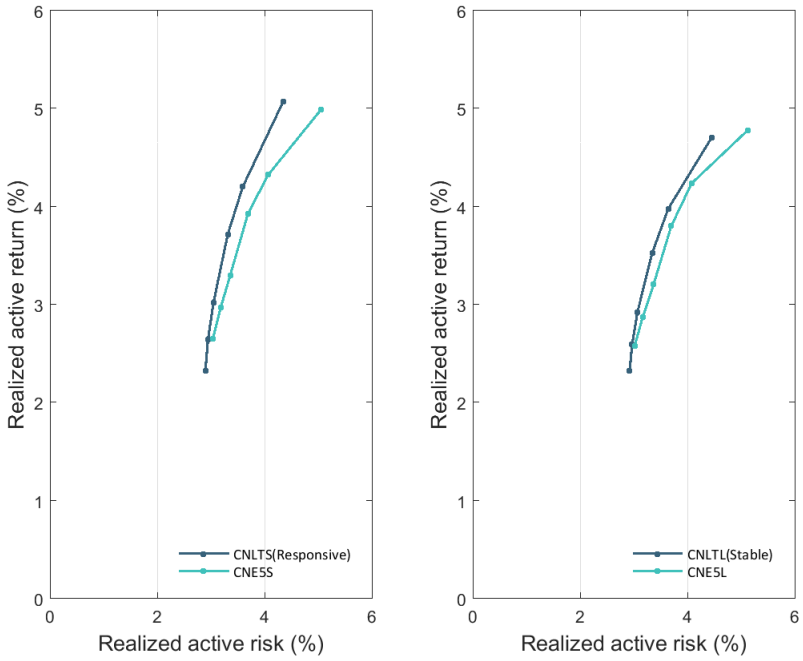
In this set of tests, the same market capitalization-weighted estimation universe of the China A Long-Term Model is used as both the investment universe and the benchmark portfolio. The alpha signal is constructed by combining the factors in the Barra China A Long-Term Model. The alpha weights are roughly by IR, but not exactly so as to avoid data mining: 10% on Momentum, Dividend Yield, Book to Price, Earnings Quality, Profitability, Long-Term reversal, respectively; and 20% on Growth and Earnings Yield.

Long-only active-strategy portfolios are constructed based on this alpha signal using the Barra China A Long-Term and the CNE5 model, respectively. The portfolios are rebalanced monthly with an 8% turnover limit. We vary the risk aversion parameter to construct the efficient frontier. The test period is July 2004 through March 2018.

In Figure 6.4, we plot the realized annual active return versus realized annual active risk for these portfolios. Transaction cost is not counted when calculating active return.

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Figure 6.4: Realized efficient frontier of active strategy before transaction costs



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7 Conclusion

This paper describes the new Barra China A Total Market Model for Long-Term Investors. The model incorporates the following key innovations:

- Alignment of factor structure and investment horizon
- New Systematic Equity Strategy factors
- Enhanced descriptor set with point-in-time data

The paper provides an empirical analysis of the Barra China A Long-Term Model. We offer a detailed presentation of the style factors, and key metrics that are reported at the individual factor level, including statistical significance, performance, volatility, and correlation with the market.

We also analyze the explanatory power of the new model, and compare it with the predecessor model, CNE5. In addition, we study the contributions to the cross-sectional dispersion from the Market factor, industry factors, and style factors. We find that each category of factors is of comparable importance in explaining the cross-sectional dispersion of equity returns.

We systematically compare the risk forecasting accuracy of the new model and the CNE5 Model. We consider several categories of portfolios, including random and factor-tilt portfolios (both long-only and dollar-neutral), and style-optimized portfolios. We find that the new model provides more accurate risk forecast for most portfolios. Finally, we observe that the new model can achieve a better performance in the analytical minimum-volatility, long-only minimum-risk, and active-strategy portfolios.

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Appendix A: Volatility Regime Adjustment

Let f_{kt} be the return of factor k on day t, and let σ_{kt} be the one-day volatility forecast for the factor at the start of the day. The standardized return of the factor is given by the ratio f_{kt}/σ_{kt}, and should have standard deviation close to 1 if the volatility forecast is accurate. As described in Appendix G, we can use the time-series standard deviation of f_{kt}/σ_{kt} to judge if the volatility forecast for this factor is unbiased across time.

Alternatively, we can use the cross-sectional standard deviation of f_{kt}/σ_{kt} to judge if the volatility forecasts for all the factors are collectively unbiased at a given point in time. We define the factor cross-sectional bias statistic B_t^F on day t as:

$$B_t^F = \sqrt{\frac{1}{K}\sum_k \left(\frac{f_{kt}}{\sigma_{kt}}\right)^2} \tag{A1}$$

where K is the total number of factors. This quantity represents an instantaneous measure of factor volatility forecast bias. For instance, if the volatility forecasts were too small on a particular day, then B_t^F would be greater than one. By observing this cross-sectional bias statistics over time, we can determine the extent to which the volatility forecasts should be adjusted to remove the biases.

We define the factor volatility multiplier λ_F as an exponentially weighted average:

$$\lambda_F = \sqrt{\sum_t (B_t^F)^2 w_t} \tag{A2}$$

where w_t is an exponential weight with Volatility Regime Adjustment half-life τ_{VRA}^F. This parameter serves as the primary determinant of model responsiveness for factor volatility. The Volatility Regime Adjustment forecasts are given by:

$$\tilde{\sigma}_k = \lambda_F \sigma_k \tag{A3}$$

This is equivalent to multiplying the entire factor covariance matrix by a single number λ_F^2. It should be clear that the Volatility Regime Adjustment has no effect on factor correlations.

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Appendix B: Optimization Bias Adjustment

Let $\mathbf{F}_0$ denote the $K \times K$ sample factor correlation matrix (FCM):

$$\mathbf{F}_0 = \mathrm{cor}(\mathbf{f}, \mathbf{f}) \tag{B1}$$

where $\mathbf{f}$ is the $K \times T$ matrix of realized factor returns, K is the number of factors, and T is the number of periods. More detail on how to estimate $\mathbf{F}_0$ is provided in Appendix H.

The sample FCM can be expressed in diagonal form as:

$$\mathbf{D}_0 = \mathbf{U}_0' \mathbf{F}_0 \mathbf{U}_0 \tag{B2}$$

where $\mathbf{U}_0$ is the $K \times K$ rotation matrix whose columns are given by the eigenvectors of $\mathbf{F}_0$. The jth element of the kth column of $\mathbf{U}_0$ gives the weight of pure factor *j* in eigenfactor *k*. The predicted variances of the eigenfactors are given by the diagonal elements of $\mathbf{D}_0$. The fact that $\mathbf{D}_0$ is diagonal indicates that the eigenfactors are mutually uncorrelated.

Although the true FCM is unobservable, we suppose for simulation purposes that the sample FCM $\mathbf{F}_0$ governs the "true" return-generating process. We generate a set of factor returns for simulation m as:

$$\mathbf{f}_m = \mathbf{U}_0 \mathbf{b}_m \tag{B3}$$

where $\mathbf{b}_m$ is a $K \times T$ matrix of simulated eigenfactor returns. The elements of row k of $\mathbf{b}_m$ are drawn from a random normal distribution with mean zero and variance given by the diagonal element $D_0(k)$ of matrix $\mathbf{D}_0$. It can be easily verified that the simulated returns in Equation B3 have a true FCM given by $\mathbf{F}_0$. Due to sampling error, however, the *estimated* FCM:

$$\mathbf{F}_m = \mathrm{cor}(\mathbf{f}_m, \mathbf{f}_m) \tag{B4}$$

will differ from the true FCM $\mathbf{F}_0$. Nevertheless, $\mathbf{F}_m$ is unbiased in the sense that $E[\mathbf{F}_m] = \mathbf{F}_0$. We diagonalize the simulated FCM:

$$\mathbf{D}_m = \mathbf{U}_m' \mathbf{F}_m \mathbf{U}_m \tag{B5}$$

where $\mathbf{U}_m$ denotes the simulated eigenfactors with estimated variances given by the diagonal elements of $\mathbf{D}_m$, i.e. $D_m(k)$.

Because we know the true distribution that governs the simulated factor returns, we can compute the true FCM of the simulated eigenfactors,

$$\tilde{\mathbf{D}}_m = \mathbf{U}_m' \mathbf{F}_0 \mathbf{U}_m \tag{B6}$$

Note that since $\mathbf{U}_m$ is not composed of the "true" eigenfactors, the matrix $\tilde{\mathbf{D}}_m$ is not diagonal. Nevertheless, our current focus is on the diagonal elements of the matrix. We compute the *simulated* volatility biases according to:

$$v(k) = \sqrt{\frac{1}{M} \sum_m \frac{\tilde{D}_m(k)}{D_m(k)}} \tag{B7}$$

where M is the total number of simulations. The simulated eigenvalue bias is computed daily, and is very stable over time.

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We now assume that the sample FCM $\mathbf{F}_0$, which uses the same correlation estimator as the simulated FCM $\mathbf{F}_m$, also suffers from the same biases. Let $\widetilde{\mathbf{D}}_0$ denote the diagonal FCM whose eigenvalues have been adjusted:

$$\widetilde{\mathbf{D}}_0 = \mathbf{v}^2\mathbf{D}_0 \tag{B8}$$

where $\mathbf{v}^2$ is a diagonal matrix whose elements are given by $v^2(k)$. The FCM in Equation B8 is now rotated from the diagonal basis to the pure factor basis using the sample eigenfactors. That is,

$$\tilde{\mathbf{F}}_0 = \mathbf{U}_0\widetilde{\mathbf{D}}_0\mathbf{U}_0' \tag{B9}$$

where $\tilde{\mathbf{F}}_0$ denotes the eigen-adjusted factor correlation matrix.

For further details, refer Menchero, Wang, and Orr (2011).

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Appendix C: Specific Risk Bayesian Shrinkage

One potential problem with using a pure time-series approach to estimate the specific volatilities for the stocks is that the specific volatilities may not fully persist out-of-sample. In particular, stocks with either extremely low or extremely high specific volatility forecasts tend to revert to the mean.

To remove this bias, we shrink our estimates toward the cap-weighted mean specific volatility of the size decile s_n to which the stock belongs. More precisely, the shrunk estimate σ_n^{SH} is given by:

$$\sigma_n^{SH} = \nu_n \bar{\sigma}(s_n) + (1 - \nu_n)\,\hat{\sigma}_n\,, \tag{C1}$$

where $\hat{\sigma}_n$ is the original forecast and ν_n is the shrinkage intensity that determines the weight given to the Bayesian prior, also known as the shrinkage target,

$$\bar{\sigma}(s_n) = \sum_{n \in S_n} w_n \hat{\sigma}_n\,, \tag{C2}$$

where w_n is the capitalization weight of stock n within the size decile. The shrinkage intensity is given by:

$$\nu_n = \frac{q|\hat{\sigma}_n - \bar{\sigma}(s_n)|}{\Delta_\sigma(s_n) + q|\hat{\sigma}_n - \bar{\sigma}(s_n)|} \tag{C3}$$

where q is an empirically determined shrinkage parameter and,

$$\Delta_\sigma(s_n) = \sqrt{\frac{1}{N(s_n)} \sum_{n \in S_n} (\hat{\sigma}_n - \bar{\sigma}(s_n))^2} \tag{C4}$$

is the standard deviation of specific risk forecasts within the size decile. The intuition behind this approach is straightforward: the more $\hat{\sigma}_n$ deviates from the mean, the greater the weight we assign to the Bayesian prior $\bar{\sigma}(s_n)$.

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Appendix D: Analysis of Style Factors and Descriptors

This appendix summarizes the results of the style factor and descriptor analysis for the new Barra China A Long-Term Model. The analysis period is July 2004 to March 2018.

The decile portfolios-performance-and-risk chart is created for each style factor:

1. The Decile plot depicts the performance of the decile portfolios. The decile portfolios are constructed as follows:
    a. On each month end, sort the stocks in the estimation universe in the ascending order of the factor exposure.
    b. Divide the stocks equally into 10 deciles. So, Decile 1 has the 10% of the stocks with the lowest factor exposure, while Decile 10 has the 10% of the stocks with the highest factor exposure.
    c. Construct an equally-weighted portfolio for each decile.

    Monthly returns for these portfolios are then computed. The annualized return is simply the sum of the entire monthly return series divided by the number of years covered by the series.
2. The Scaled Factor Return plot depicts the performance of the style factor from both the *All Styles* and *Single Style* monthly regressions. As the names suggest, the two regressions differ in the style factors used in them: all the style factors are used in the *All Styles* regression, while only this one style factor is used in the *Single Style* regression. The Market factor and all the industry factors are used in both regressions. The *All Styles* monthly regression is the same one used to generate the official monthly factor returns of the Barra China A Long-Term Model. Both return series are scaled to have an annual volatility of 1%, and are then used to generate the performance lines by a simple cumulative sum.
3. The Daily Factor Return plot depicts:
    a. The cumulative daily factor return, where the daily factor returns are the official ones from the Barra China A Long-Term Model, generated by the *All Styles* daily regressions, and the cumulative return is a simple cumulative sum of the daily factor returns.
    b. The maximum drawdown in the cumulative daily factor return.
    c. The Z score of the weekly factor return, computed as:

    $$z_w(t) = \frac{1}{\sqrt{5}} \sum_{l=0}^{4} z_d(t-l)$$

    where $z_d(t)$ is the daily Z score computed as the daily factor return divided by a daily volatility estimated from a 126-day window lagged by 10 days.
4. The Factor Risk plot depicts factor risk forecast.

We also report the summary statistics for each factor in a table. All the columns except for the is $CV\ R^2$ Gain and the $Studentized\ R^2$ Gain in the table are generated from the official monthly factor returns of the Barra China A Long-Term Model. The $CV\ R^2$ gain is generated by comparing the $CV\ R^2$ for the *All Styles* regression used to generate the official monthly factor returns, and a second one from which the style factor is left out. The

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difference between the two $CV\ R^2$s can be interpreted as the net contribution by this style factor. The $Studentized\ R^2$ Gain is obtained similarly.

For the styles with two or more descriptors, we also conduct a what-if study of the performance of each descriptor by replacing the factor with this descriptor, in the *All Styles* monthly regression. The summary statistics for the descriptors are reported in the same table with the factor. In addition, we also plot the cumulative monthly returns of the descriptors, and report the correlation of the monthly returns of the descriptors and the factor.

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Beta

Description	Captures the market risk that cannot be explained by the Country factor.
Motivation	The Capital Asset Pricing Model (CAPM), one of the workhorses of finance theory, describes the relationship between risk and expected return. One of the implications of the model is that one of the key determinants of an investor's required rate of return and stock risk is stock beta.
Start Date	Dec 31, 1998
Frequency	Daily
Exposure Interpretation	A positive exposure indicates a high beta stock. A negative exposure indicates a low beta stock.
Descriptors	• Historical Beta

Figure D.1: Decile portfolios, performance, and risk of the Beta factor

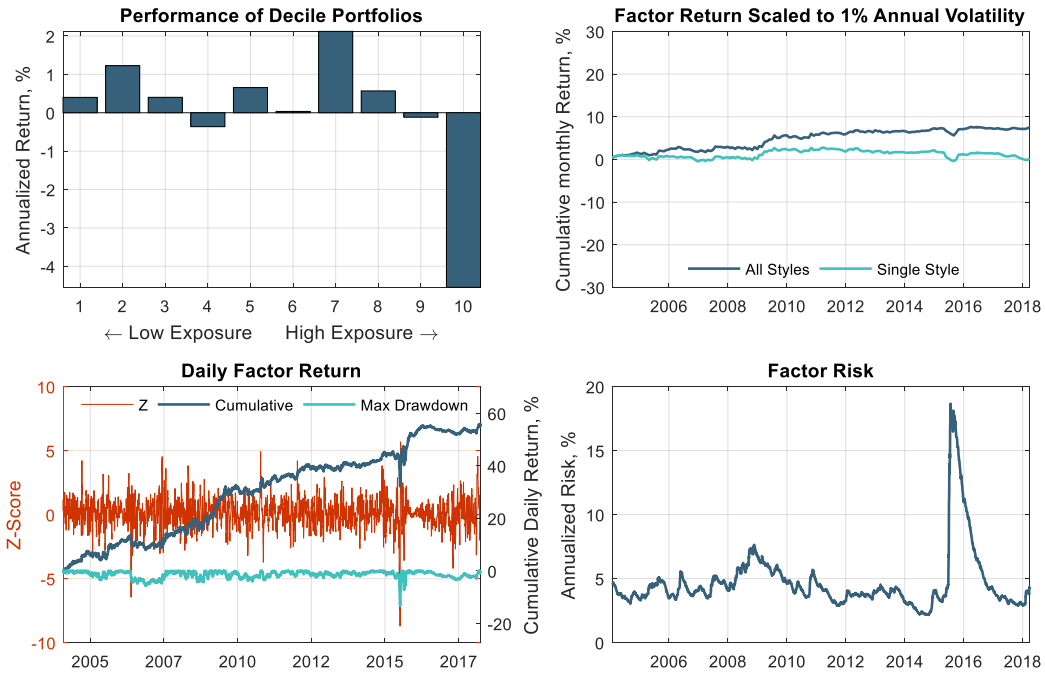


Table D.1: Summary statistics for the Beta factor

	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bps)	Maximum Drawdown (%)	Corr. with Market
Beta	3.02	57.9	2.48	4.66	0.53	39.82	43.38	7.96	0.54

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Book-to-Price

Description	Explains the return component attributable to a stock's book-to-price ratio and can be an indicator of value. This factor is based on Systematic Equity Strategies.
Motivation	Traditional value indicator, part of the influential Fama-French Three-Factor Model. Historically, high-value companies earned higher returns and experienced higher risk relative to low-value companies.
Start Date	Mar 31, 2004
Frequency	Daily
Exposure Interpretation	A positive exposure indicates an undervalued stock. A negative exposure indicates an overvalued stock.
Descriptors	• Book-to-Price

Figure D.2: Decile portfolios, performance, and risk of the Book-to-Price factor

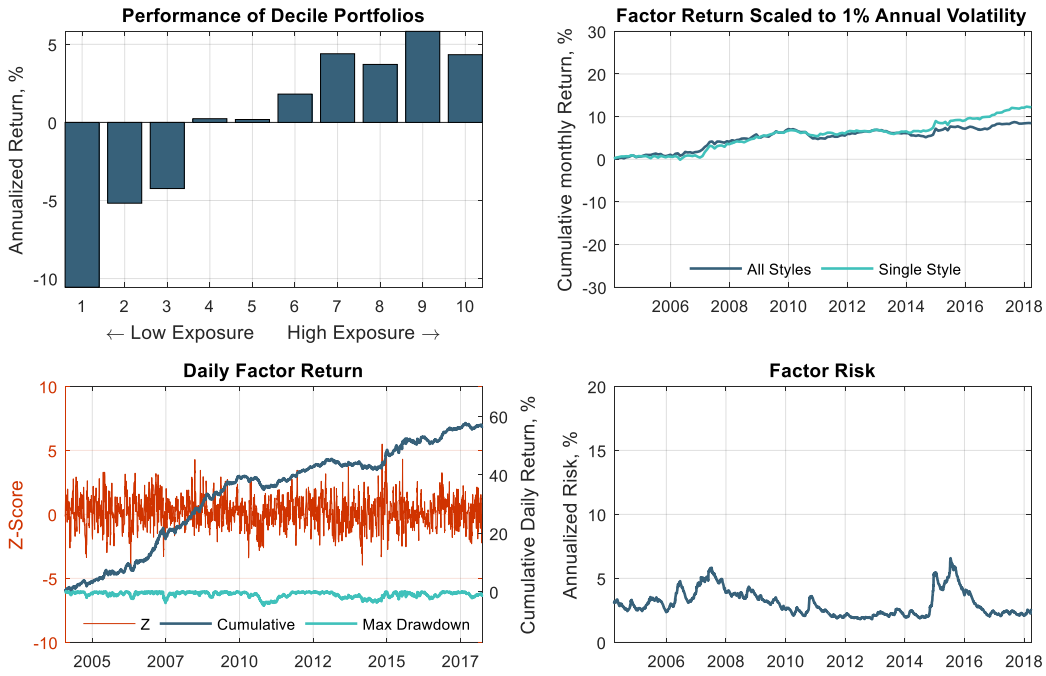


Table D.2: Summary statistics for the Book-to-Price factor

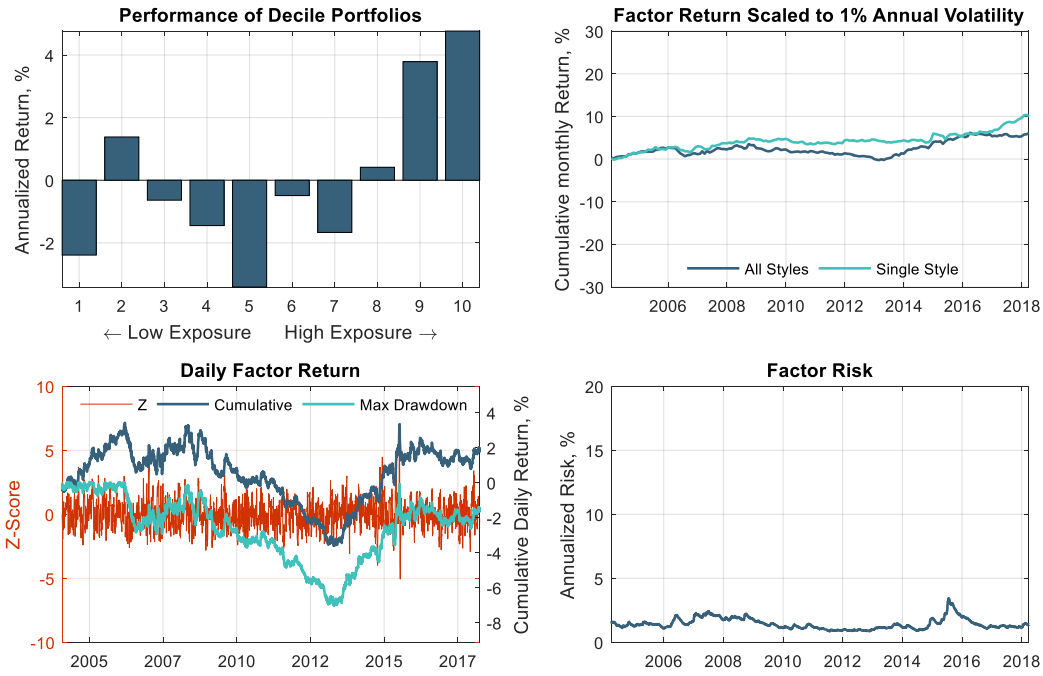
	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bp)	Maximum Drawdown (%)	Corr. with Market
Book-to-Price	1.69	34.5	1.90	3.20	0.59	11.21	14.93	7.53	0.14

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Dividend Yield

Description	Captures differences in stock returns attributable to stock's historical and predicted dividend-to-price ratios. This factor is based on Systematic Equity Strategies.
Motivation	Dividend is one of the central inputs in Gordon Growth Model for valuing a company's stock price. Rather than building a dividend-based structural model for valuation, we apply a relative valuation approach and use the Dividend Yield factor to capture common variation and risk differences between dividend paying companies.
Start Date	Mar 31, 2004
Frequency	Daily
Exposure Interpretation	A positive exposure indicates a high historical/predicted dividend yield. A negative exposure indicates a low historical/predicted dividend yield.
Descriptors	• (Trailing) Dividend-to-Price • Analyst-Predicted Dividend-to-Price

Figure D.3: Decile portfolios, performance, and risk of the Dividend Yield factor



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Table D.3: Summary statistics for the Dividend Yield descriptors and factor

	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bp)	Maximum Drawdown (%)	Corr. with Market
Dividend-to-Price	0.92	6.0	0.40	1.34	0.30	-3.34	1.06	4.71	0.01
Analyst-Predicted Dividend-to-Price	1.13	16.3	0.11	1.42	0.08	-1.61	2.73	4.17	0.11
Dividend Yield	0.94	9.9	0.60	1.40	0.43	-3.11	1.31	5.25	0.04

Figure D.4: Cumulative monthly returns of the Dividend Yield descriptors

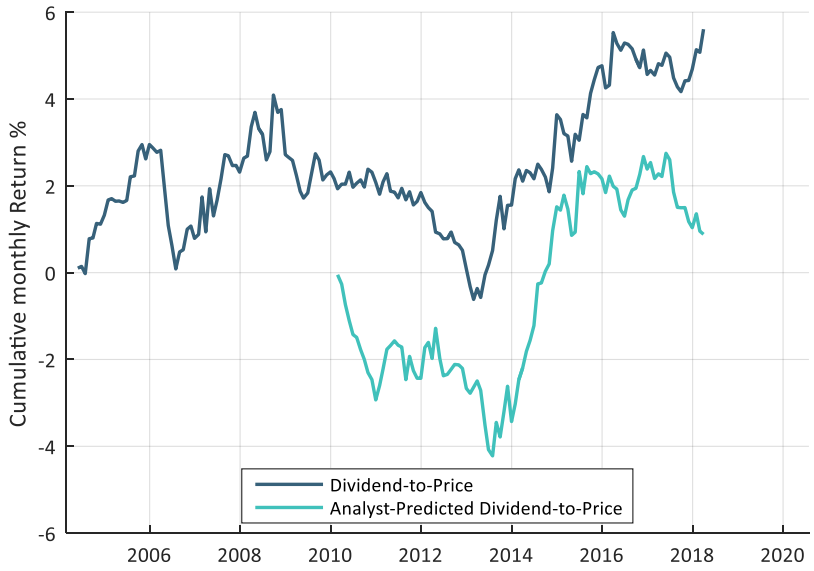


Table D.4: Correlation of the monthly returns of the Dividend Yield descriptors and factor

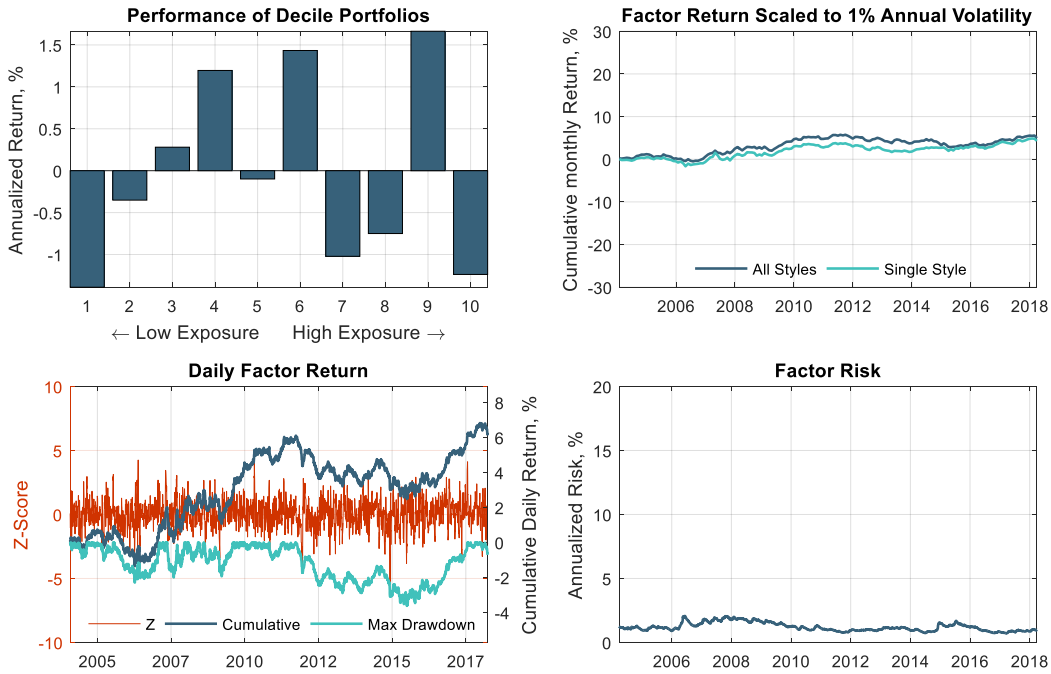
	(Trailing) Dividend-to-Price	Analyst-Predicted Dividend-to-Price	Dividend Yield
(Trailing) Dividend-to-Price	1		
Analyst-Predicted Dividend-to-Price	0.65	1	
Dividend Yield	0.97	0.8	1

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Earnings Quality

Description	Explains stock return differences due to the accrual components of their earnings. This factor is based on Systematic Equity Strategies.
Motivation	Sloan R. (1996) illustrates the importance of distinguishing persistent and non-persistent (accruals) components of company earnings in valuing companies. The accrual components of earnings involve significant management discretion and are more prone to manipulations. Hence, earnings growth driven by a large accrual component is seen as less "sustainable", or of "low quality".
Start Date	Mar 31, 2004
Frequency	Daily
Exposure Interpretation	A positive exposure indicates low accruals and low uncertainty around firm fundamentals. A negative exposure indicates high accruals and high uncertainty around firm fundamentals.
Descriptors	• Accruals - Balance Sheet Version • Accruals - Cash flow Version

Figure D.5: Decile portfolios, performance, and risk of the Earnings Quality factor

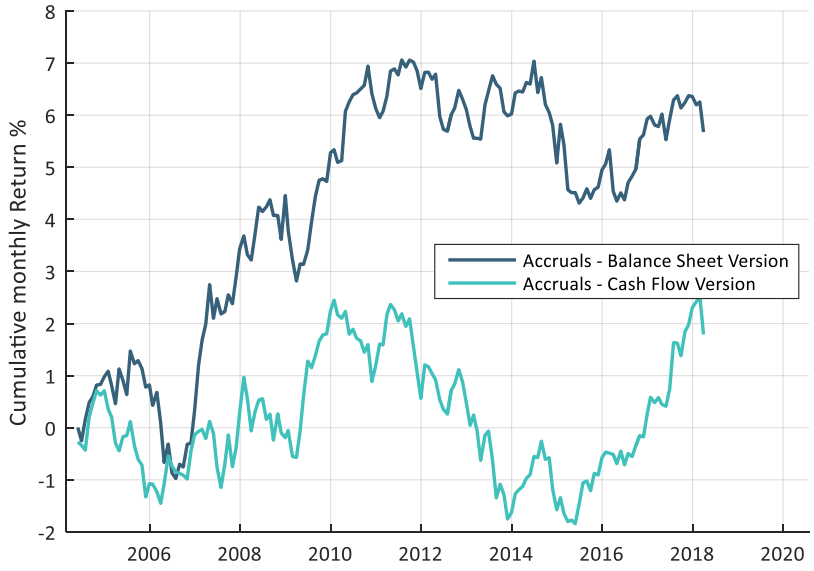


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Table D.5: Summary statistics for the Earnings Quality descriptors and factor

	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bp)	Maximum Drawdown (%)	Corr. with Market
Accruals - Balance Sheet Version	1.04	12.0	0.41	1.24	0.33	-2.64	2.16	2.75	-0.07
Accruals - Cash Flow Version	1.12	15.6	0.13	1.17	0.11	-2.59	2.73	4.29	0.03
Earnings Quality	1.14	15.8	0.43	1.25	0.35	-2.05	2.98	3.61	-0.02

Figure D.6: Cumulative monthly returns of the Earnings Quality descriptors



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Table D.6: Correlation of the monthly returns of the Earnings Quality descriptors and factor

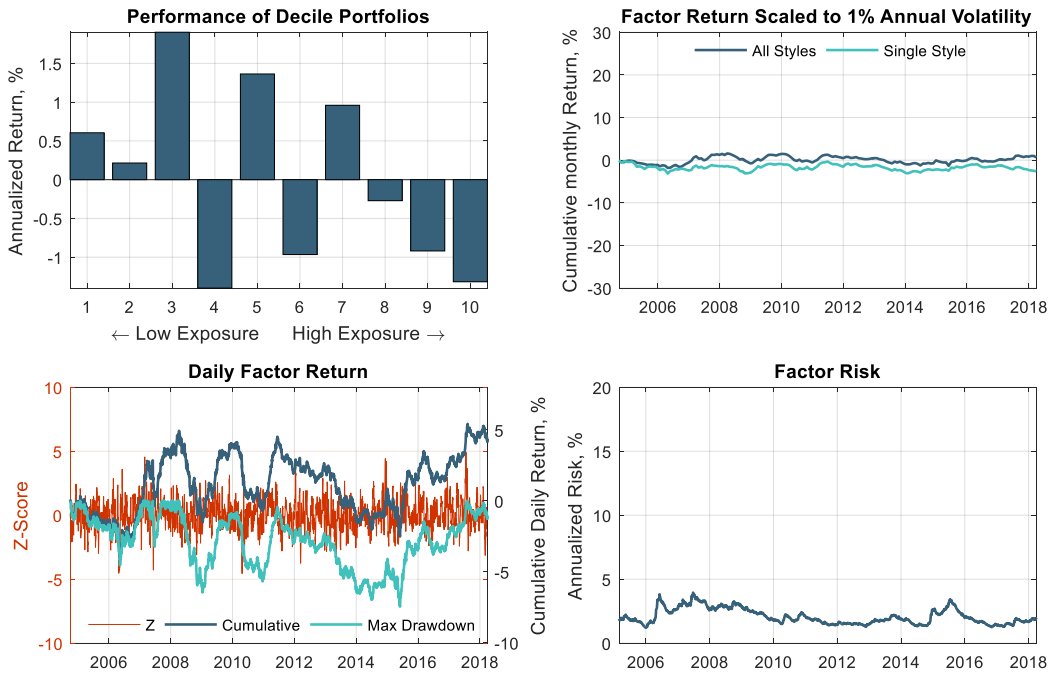
	Accruals - Balance Sheet Version	Accruals - Cash Flow Version	Earnings Quality
Accruals - Balance Sheet Version	1		
Accruals - Cash Flow Version	0.41	1	
Earnings Quality	0.89	0.78	1

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Earnings Variability

Description	Explains stock return differences due to variability in earnings, sales, and cash-flow, and also in analysts' predicted earnings-to-price.
Motivation	Companies with higher earnings quality have less uncertainty around their fundamentals and tend to outperform the market average over the long run.
Start Date	Feb 28, 2005
Frequency	Daily
Exposure Interpretation	A positive exposure indicates high uncertainty around company fundamentals. A negative exposure indicates low uncertainty around firm fundamentals.
Descriptors	• Standard Deviation of Analyst Forecast Earnings-to-Price • Variability in Earnings • Variability in Cash flows • Variability in Sales

Figure D.7: Decile portfolios, performance, and risk of the Earnings Variability factor

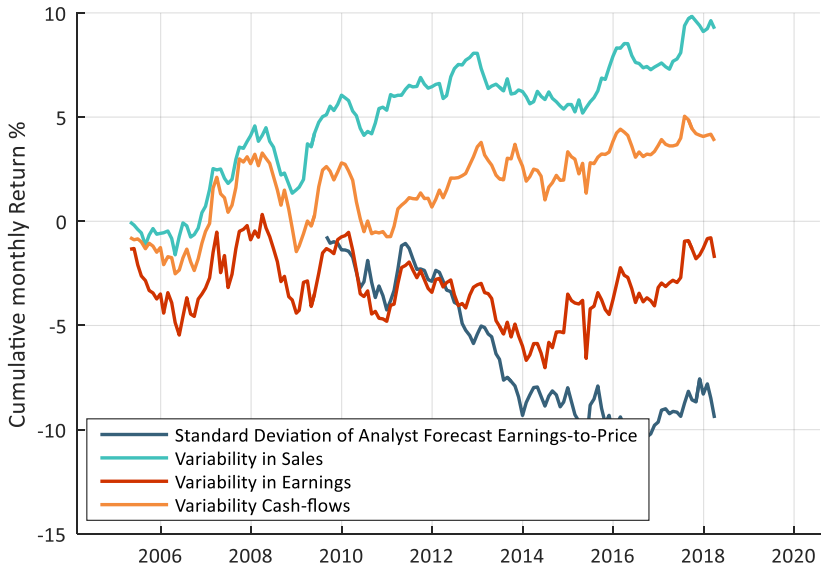


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Table D.7: Summary statistics for the Earnings Variability descriptors and factor

	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bp)	Maximum Drawdown (%)	Corr. with Market
Standard Deviation of Analyst Forecast Earnings-to-Price	1.42	25.0	-1.09	1.93	-0.56	2.47	6.61	10.04	0.07
Variability in Sales	1.01	7.7	0.71	1.44	0.49	-3.15	1.75	3.23	0.22
Variability in Earnings	1.57	34.0	-0.14	2.43	-0.06	5.66	10.11	7.36	0.06
Variability Cash-flows	1.32	21.8	0.30	1.84	0.16	0.79	5.47	4.73	0.19
Earnings Variability	1.67	35.0	0.11	2.27	0.05	6.38	10.95	6.62	0.16

Figure D.8: Cumulative monthly returns of the Earnings Variability descriptors



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Table D.8: Correlation of the monthly returns of the Earnings Variability descriptors and factor

	Standard Deviation of Analyst Forecast Earnings-to-Price	Variability in Sales	Variability in Earnings	Variability Cash-flows	Earnings Variability
Standard Deviation of Analyst Forecast Earnings-to-Price	1				
Variability in Sales	0.06	1			
Variability in Earnings	0.15	0.17	1		
Variability Cash-flows	0.11	0.44	0.58	1	
Earnings Variability	0.52	0.61	0.76	0.81	1

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Earnings Yield

Description	Describes stock return differences due to various ratios of the company's earnings relative to its price. This factor is based on Systematic Equity Strategies.
Motivation	The Earnings Yield factor is one of the relative valuation multiples popular in the finance industry. Price multiples characterize a stock's relative "market" valuation and differ from multiples scaled by other metrics such as assets, sales, or book value. Most company valuations used by the finance industry are relative valuations based on some company multiples and comparables. It is pointed out in academic literature that almost 85% of equity research reports and more than 50% of all acquisitions are based upon a company multiple.
Start Date	Mar 31, 2004
Frequency	Daily
Exposure Interpretation	A positive exposure indicates a high historical/predicted earnings yield ('cheap' stocks). A negative exposure indicates a low historical/predicted earnings yield ('expensive' stocks).
Descriptors	• Cash Earnings-to-Price • Enterprise Multiple (EBIT to EV) • Earnings-to-Price • Analyst-Predicted Earnings-to-Price

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Figure D.9: Decile portfolios, performance, and risk of the Earnings Yield factor

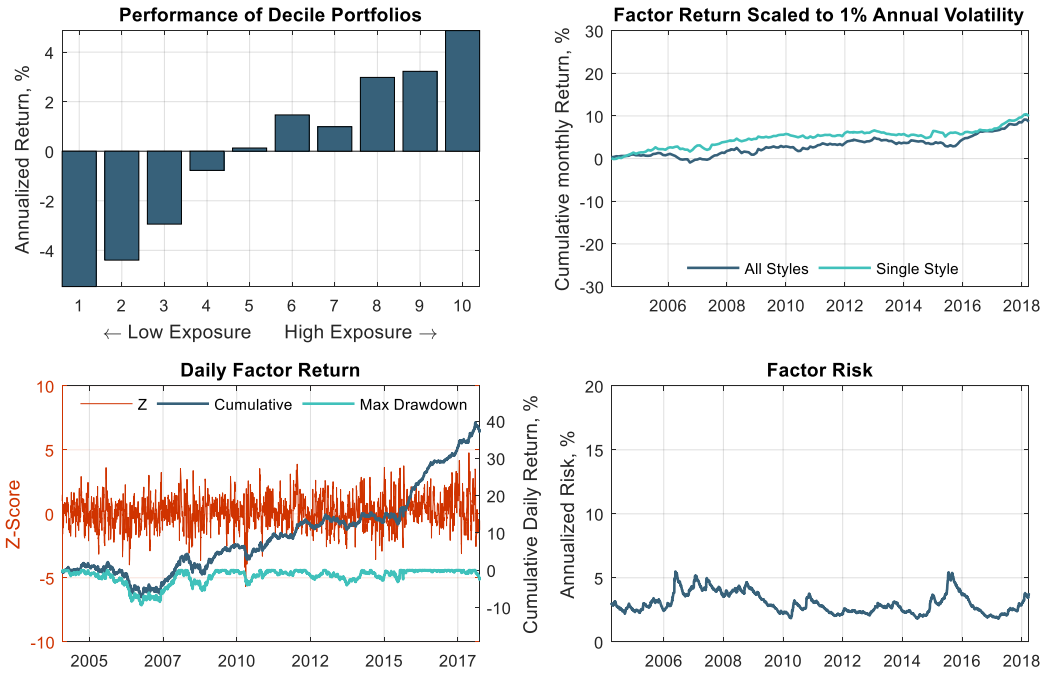


Table D.9: Summary statistics for the Earnings Yield descriptors and factor

	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bp)	Maximum Drawdown (%)	Corr. with Market
Cash-Earnings-to-Price	1.43	25.1	-0.28	2.64	-0.11	3.16	7.46	16.14	0.29
Earnings-to-Price	1.75	35.9	1.32	2.98	0.44	8.68	12.78	7.29	0.24
Analyst-Predicted Earnings-to-Price	1.82	36.8	1.42	2.77	0.51	11.10	14.85	4.89	0.26
Enterprise Multiple (EBIT to EV)	1.25	21.6	-0.26	2.17	-0.12	0.17	4.61	12.79	0.32
Earnings Yield	1.75	29.2	1.94	3.20	0.61	9.51	13.54	7.18	0.26

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Figure D.10: Cumulative monthly returns of the Earnings Yield descriptors

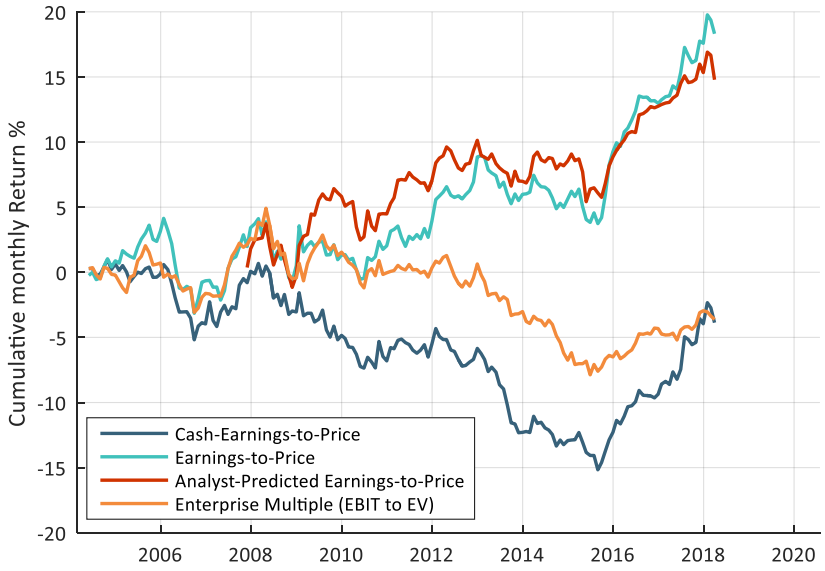


Table D.10: Correlation of the monthly returns of the Earnings Yield descriptors and factor

	Cash-Earnings-to-Price	Earnings-to-Price	Analyst-Predicted Earnings-to-Price	Enterprise Multiple (EBIT to EV)	Earnings Yield
Cash-Earnings-to-Price	1				
Earnings-to-Price	0.77	1			
Analyst-Predicted Earnings-to-Price	0.63	0.73	1		
Enterprise Multiple (EBIT to EV)	0.72	0.74	0.58	1	
Earnings Yield	0.84	0.91	0.89	0.85	1

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Growth

Description	Measures company growth prospects using historical sales growth and historical and predicted earnings growth. This factor is based on Systematic Equity Strategies.
Motivation	Growth is one of the factors that determine the future cash flow and dividends paid out to investors and, thus, future stock prices. The Gordon Growth Model, which is an example of the Dividend Discount Model (DDM), for valuing a company's stock price is based on the net present value of future dividends. The model predicts the relationship between stock price, future dividends, cost of capital and the dividend growth rate. We use the Growth factor in the risk model as a proxy for future dividend growth rate.
Start Date	Jan 31, 2005
Frequency	Daily
Exposure Interpretation	A positive exposure indicates a high historical/predicted growth. A negative exposure indicates a low historical/predicted growth.
Descriptors	• Analyst Predicted Earnings Long-Term Growth • Earnings Per Share Growth Rate • Sales Per Share Growth Rate

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Figure D.11: Decile portfolios, performance, and risk of the Growth factor

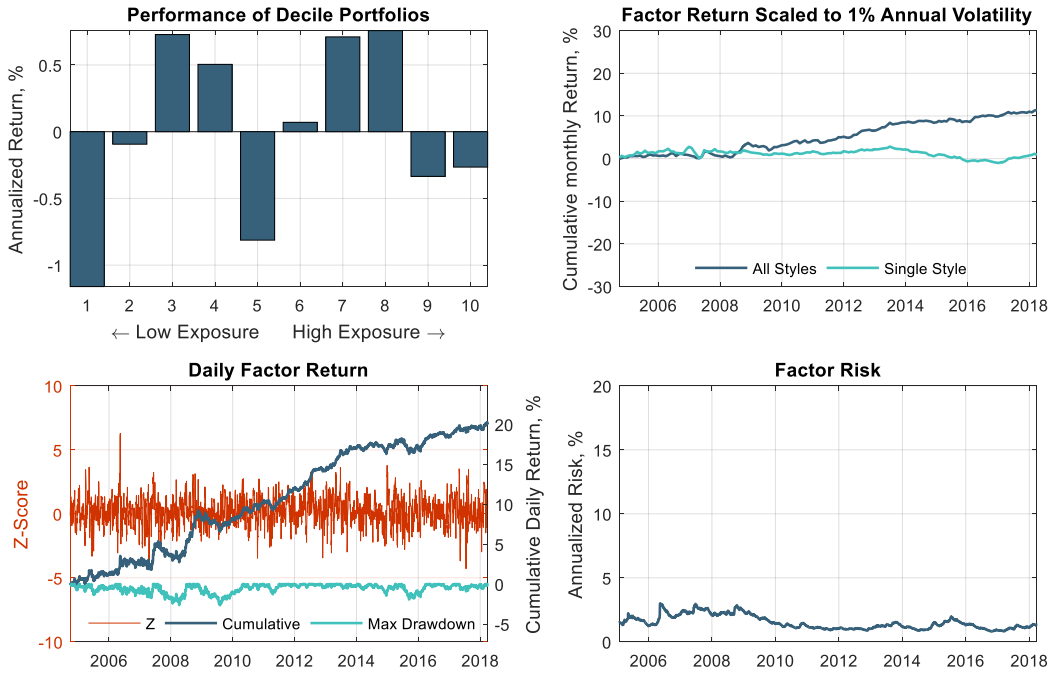


Table D.11: Summary statistics for the Growth descriptors and factor

	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bp)	Maximum Drawdown (%)	Corr. with Market
Earnings per Share Growth Rate	0.99	12.7	-0.07	1.61	-0.04	-3.79	0.99	7.13	-0.14
Sales per Share Growth Rate	1.08	12.7	1.00	1.46	0.69	-3.05	2.04	2.91	-0.25
Analyst-Predicted Earnings Long-term Growth	1.36	26.0	3.35	2.35	1.43	-0.05	4.98	3.69	0.04
Growth	1.03	12.9	1.13	1.35	0.84	-3.82	1.27	2.39	-0.23

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Figure D.12: Cumulative monthly returns of the Growth descriptors

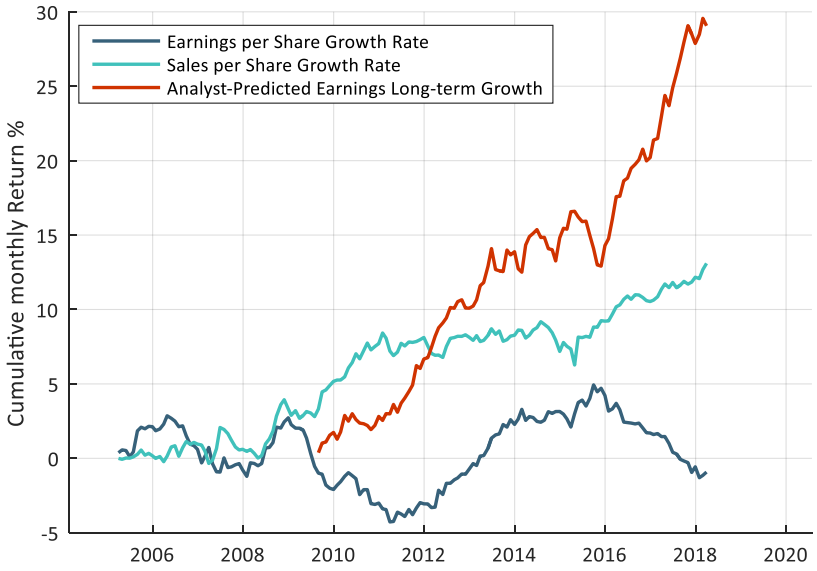


Table D.12: Correlation of the monthly returns of the Growth descriptors and factor

	Earnings per Share Growth Rate	Sales per Share Growth Rate	Analyst-Predicted Earnings Long-term Growth	Growth
Earnings per Share Growth Rate	1			
Sales per Share Growth Rate	0.41	1		
Analyst-Predicted Earnings Long-term Growth	-0.06	0.01	1	
Growth	0.62	0.84	0.62	1

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Investment Quality

Description	Measures the tendency of management to pursue empire-building and over-invest and management views when the stock is over/under priced.
Motivation	There is evidence that companies with corporate events associated with asset expansion (that is, acquisitions, public equity offering, etc.) tend to experience lower returns than companies with corporate events associated with asset contraction (that is, spinoff, share repurchase, etc.). These findings suggest that investors may have a bias in the capitalization of company asset investments and disinvestment decisions. Related to these findings, there is evidence that management tends to issue or repurchase shares when the company is overvalued or undervalued, and investors underreact to that information. Also, high capital expenditures and asset growth are associated with the phenomenon of "empire-building" that has a negative impact on company future performance.
Start Date	Jan 31, 2005
Frequency	Daily
Exposure Interpretation	A positive exposure indicates low asset and capital expenditure growth and low net equity issuance. A negative exposure indicates high asset and capital expenditure growth and high net equity issuance.
Descriptors	• Total Assets Growth Rate • Capital Expenditure Growth • Issuance Growth

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Figure D.13: Decile portfolios, performance, and risk of the Investment Quality factor

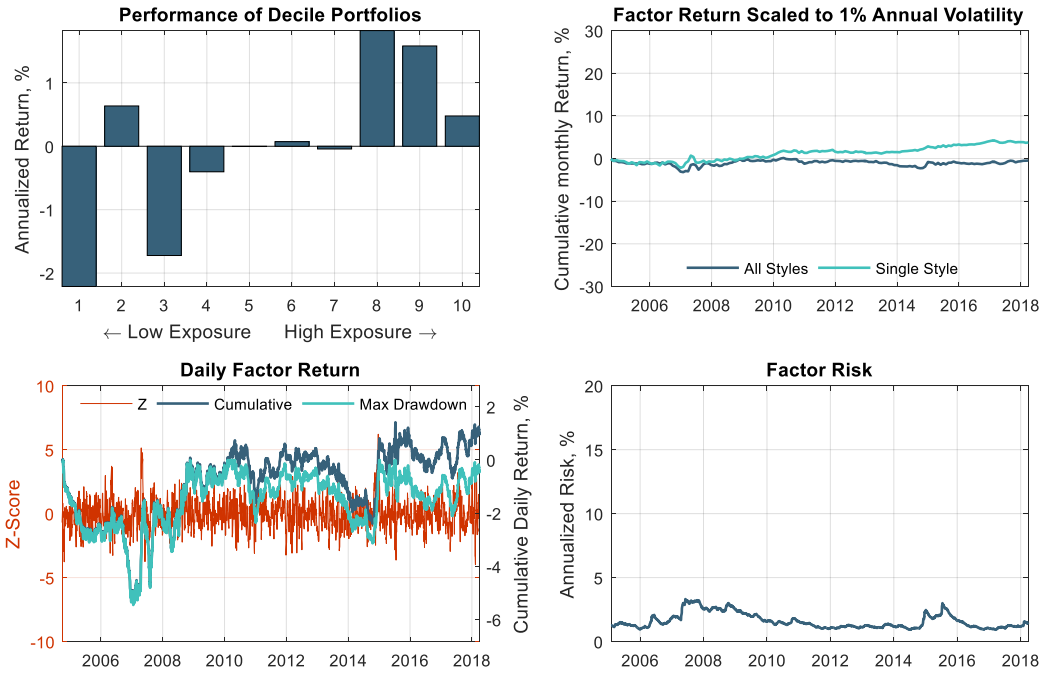


Table D.13: Summary statistics for the Investment Quality descriptors and factor

	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bp)	Maximum Drawdown (%)	Corr. with Market
Total Assets Growth Rate	1.36	26.1	-0.54	2.19	-0.25	0.91	5.86	9.00	0.07
Issuance Growth	1.09	16.6	0.14	1.56	0.09	-2.57	2.48	4.42	-0.03
Capital Expenditure Growth	0.98	8.9	0.32	1.37	0.23	-2.88	1.71	4.46	0.10
Investment Quality	1.20	19.6	-0.06	1.84	-0.03	-0.26	4.61	5.76	0.06

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Figure D.14: Cumulative monthly return of the Investment Quality descriptors

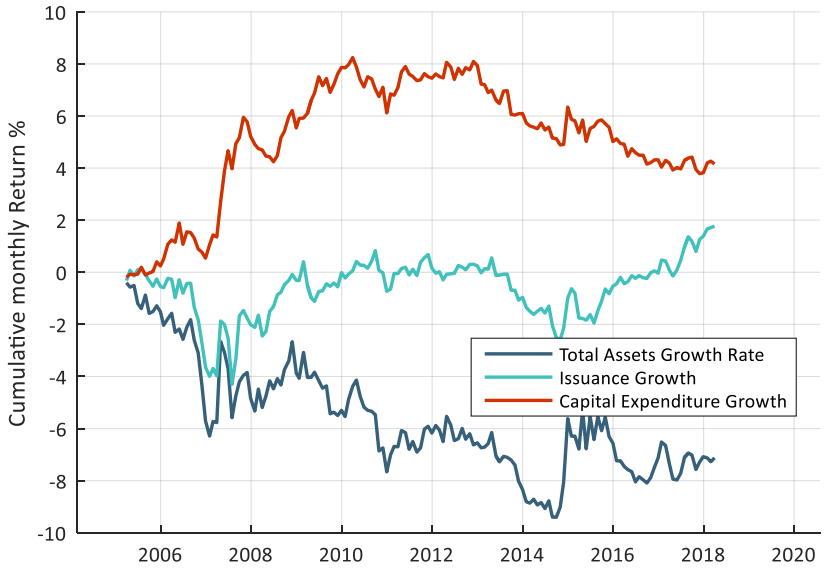


Table D.14: Correlation of the monthly returns of the Investment Quality descriptors and factor

	Total Assets Growth Rate	Issuance Growth	Capital Expenditure Growth	Investment Quality
Total Assets Growth Rate	1			
Issuance Growth	0.54	1		
Capital Expenditure Growth	0.48	0.25	1	
Investment Quality	0.88	0.83	0.61	1

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Leverage

Description	Captures common variation in stock returns due to differences in the level of company leverage.
Motivation	We view highly-leveraged firms as riskier because they cannot change their production easily. In particular, they cannot scale down production during recessionary periods without increasing the probability of default. Highly-leveraged firms are saddled with too much capital during periods of low productivity and, relative to the low-leveraged firms, are more sensitive to interest rate shocks. See Bhandari (1988) for more discussions.
Start Date	Mar 31, 2004
Frequency	Daily
Exposure Interpretation	A positive exposure indicates a high leverage. A negative exposure indicates a low leverage.
Descriptors	• Debt-to-Assets • Book Leverage • Market Leverage

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Figure D.15: Decile portfolios, performance, and risk of the Leverage factor

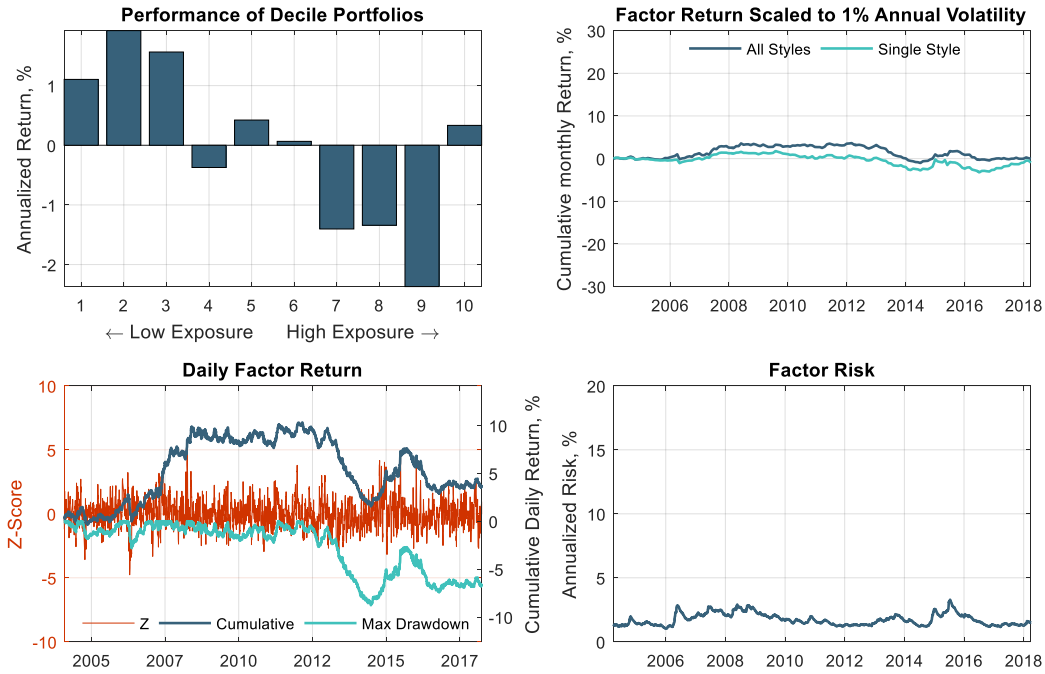


Table D.15: Summary statistics for the Leverage descriptors and factor

	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bp)	Maximum Drawdown (%)	Corr. with Market
Debt-to-Assets	1.22	18.6	0.14	1.85	0.08	0.87	5.25	8.03	-0.04
Book Leverage	1.10	13.2	-0.07	1.49	-0.05	-1.88	2.77	6.85	-0.03
Market Leverage	1.16	18.6	-0.24	1.72	-0.14	-0.35	3.82	9.26	-0.00
Leverage	1.24	20.5	-0.00	1.92	-0.00	1.49	5.73	8.96	-0.02

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Figure D.16: Cumulative monthly return of the Leverage descriptors

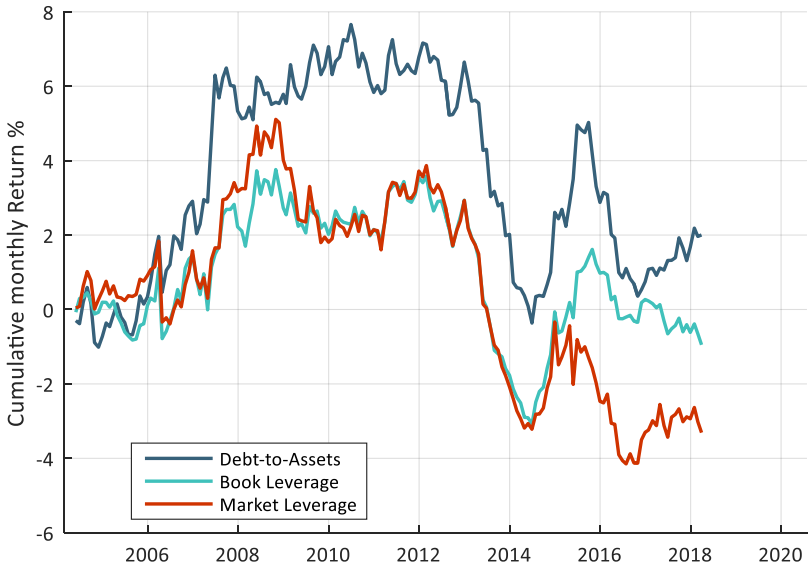


Table D.16: Correlation of the monthly returns of the Leverage descriptors and factor

	Debt-to-Assets	Book Leverage	Market Leverage	Leverage
Debt-to-Assets	1			
Book Leverage	0.53	1		
Market Leverage	0.47	0.89	1	
Leverage	0.81	0.91	0.88	1

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Liquidity

Description	Captures common variations in stock returns due to the amount of relative trading and differences in the impact of trading on stock returns.
Motivation	The ability of an investor to convert stock holdings into cash determines an illiquidity premium, that is, excess returns that investors require for holding difficult-to-sell illiquid stocks. The risk of holding illiquid stocks is that an investor may not be able to sell her holdings without incurring significant losses when she needs to raise cash. While volatility of highly-liquid stocks is primarily driven by changes in company fundamentals, volatility of illiquid stocks may be driven by company fundamentals as well as the investor's needs to raise cash by liquidating illiquid positions.
Start Date	Dec 31, 1998
Frequency	Daily
Exposure Interpretation	A positive exposure indicates a high liquidity. A negative exposure indicates a low liquidity.
Descriptors	• Monthly Share Turnover • Quarterly Share Turnover • Annual Share Turnover • Annual Traded Value Ratio

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Figure D.17: Decile portfolios, performance, and risk of the Liquidity factor

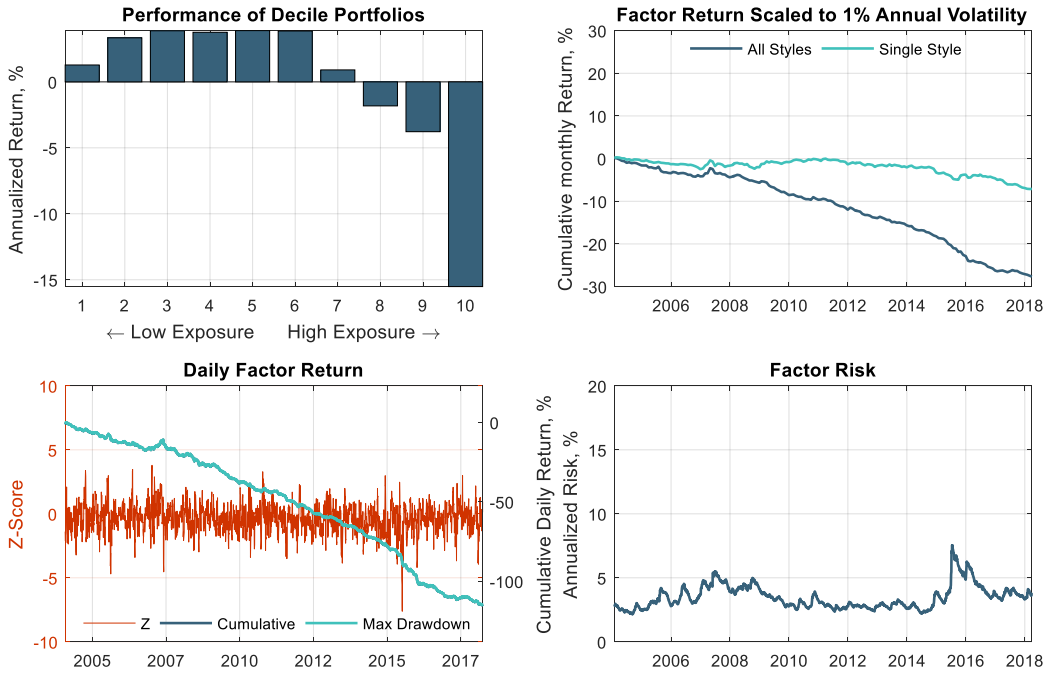


Table D.17: Summary statistics for the Liquidity descriptors and factor

	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bp)	Maximum Drawdown (%)	Corr. with Market
Monthly Share Turnover	3.03	62.6	-8.20	3.62	-2.27	40.05	44.39	116.89	0.33
Quarterly Share Turnover	2.63	56.1	-6.59	3.48	-1.90	27.05	31.67	94.51	0.35
Annual Share Turnover	1.99	39.2	-3.20	2.89	-1.11	13.65	18.27	47.90	0.25
Annual Traded Value Ratio	2.19	45.6	-4.59	3.28	-1.40	17.49	22.18	66.10	0.31
Liquidity	2.61	53.2	-6.78	3.49	-1.94	26.95	31.54	97.09	0.33

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Figure D.18: Cumulative monthly return of the Liquidity descriptors

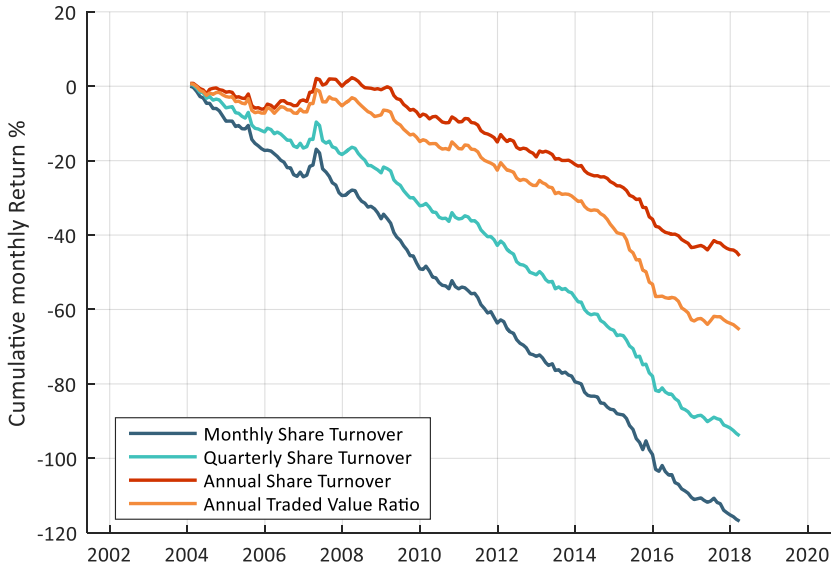


Table D.18: Correlation of the monthly returns of the Liquidity descriptors and factor

	Monthly Share Turnover	Quarterly Share Turnover	Annual Share Turnover	Annual Traded Value Ratio	Liquidity
Monthly Share Turnover	1				
Quarterly Share Turnover	0.91	1			
Annual Share Turnover	0.79	0.87	1		
Annual Traded Value Ratio	0.83	0.94	0.97	1	
Liquidity	0.92	0.97	0.95	0.98	1

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Long-Term Reversal

Description	Explains common variation in returns related to a long-term (five years ex. recent thirteen months) stock price behavior. This factor is based on Systematic Equity Strategies.
Motivation	The early evidence for long-term reversal phenomena in stock returns goes back to De Bondt and Thaler (1985). Our own research illustrates that actively managed US mutual funds have significant exposure to this factor.
Start Date	Dec 31, 1998
Frequency	Daily
Exposure Interpretation	A positive exposure indicates a low long-term momentum (poor long-term performance ex. recent performance). A negative exposure indicates a high long-term momentum (good long-term performance ex. recent performance).
Descriptors	• Long-Term Relative Strength • Long-Term Historical Alpha

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Figure D.19: Decile portfolios, performance, and risk of the Long-Term Reversal factor

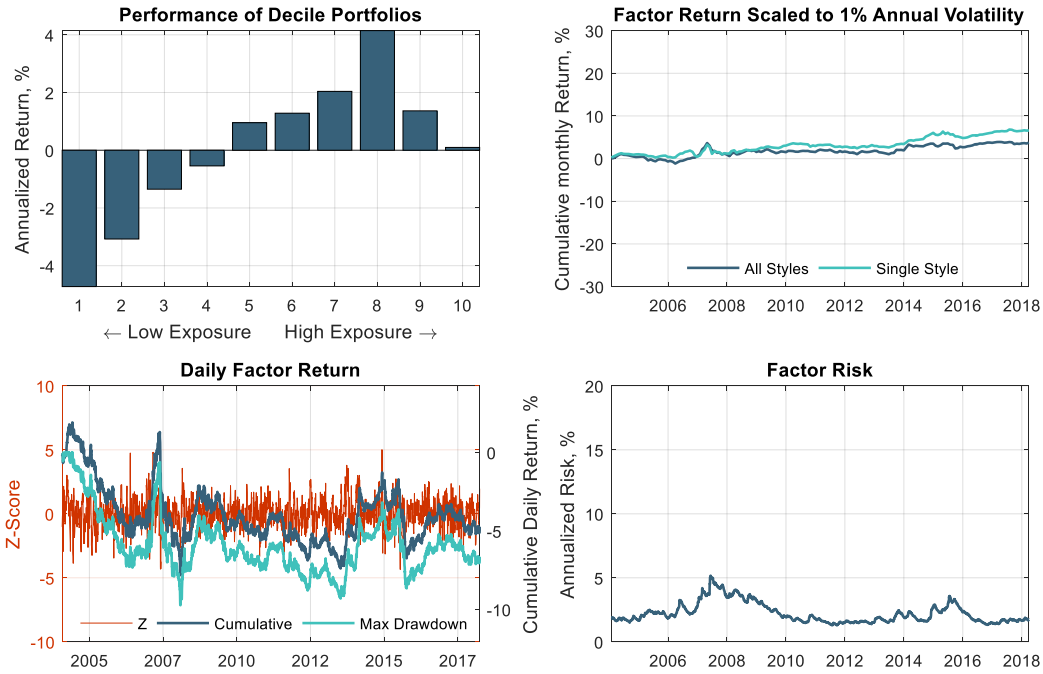


Table D.19: Summary statistics for the Long-Term Reversal descriptors and factor

	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bp)	Maximum Drawdown (%)	Corr. with Market
Long-Term Relative Strength	1.44	22.8	0.58	2.31	0.25	3.18	8.14	6.31	0.19
Long-Term Historical Alpha	1.40	26.3	0.68	2.30	0.30	3.98	8.79	6.90	0.19
Long-Term Reversal	1.43	24.6	0.60	2.29	0.26	3.92	8.78	7.07	0.19

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Figure D.20: Cumulative monthly return of the Long-Term Reversal descriptors

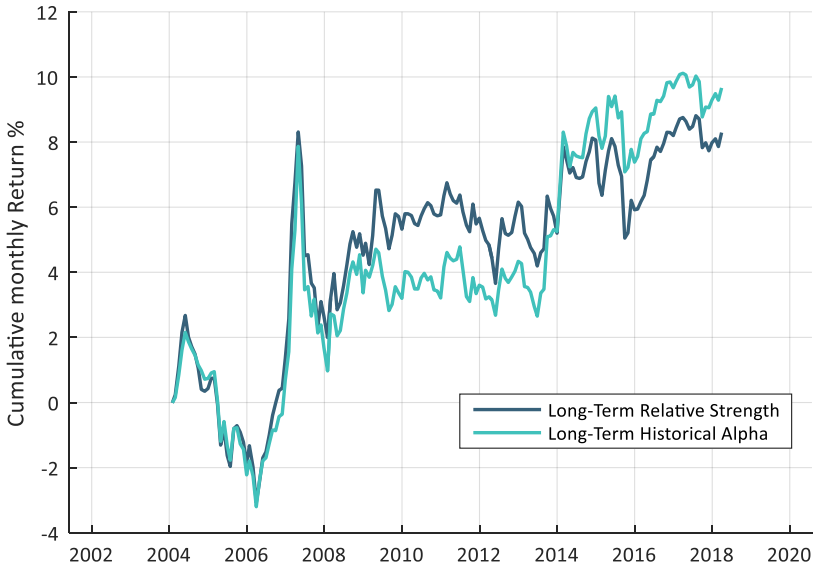


Table D.20: Correlation of the monthly returns of the Long-Term Reversal descriptors and factor

	Long-Term Relative Strength	Long-Term Historical Alpha	Long-Term Reversal
Long-Term Relative Strength	1		
Long-Term Historical Alpha	0.90	1	
Long-Term Reversal	0.94	0.93	1

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Mid Capitalization

Description	Captures deviations from linearity in the relationship between returns and the logarithm of market capitalization (Size factor). This factor explains differences in risk and return for mid-capitalization stocks from small-cap and large-cap stocks.
Motivation	A closer look at the relationship between company stock returns or risk and the company log of market capitalization reveals deviations from a linear relationship. In particular, a change in expected returns and risk tends to increase more rapidly than implied by a linear model as we move from large-capitalization companies to small-capitalization companies. To capture this non-linear relationship in a linear factor framework, we introduced the Mid Capitalization factor.
Start Date	Dec 31, 1998
Frequency	Daily
Exposure Interpretation	A positive exposure indicates mid capitalization. A negative exposure indicates the large and small capitalization.
Descriptors	• Cube of Size Exposure

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Figure D.21: Decile portfolios, performance, and risk of the Mid Capitalization factor

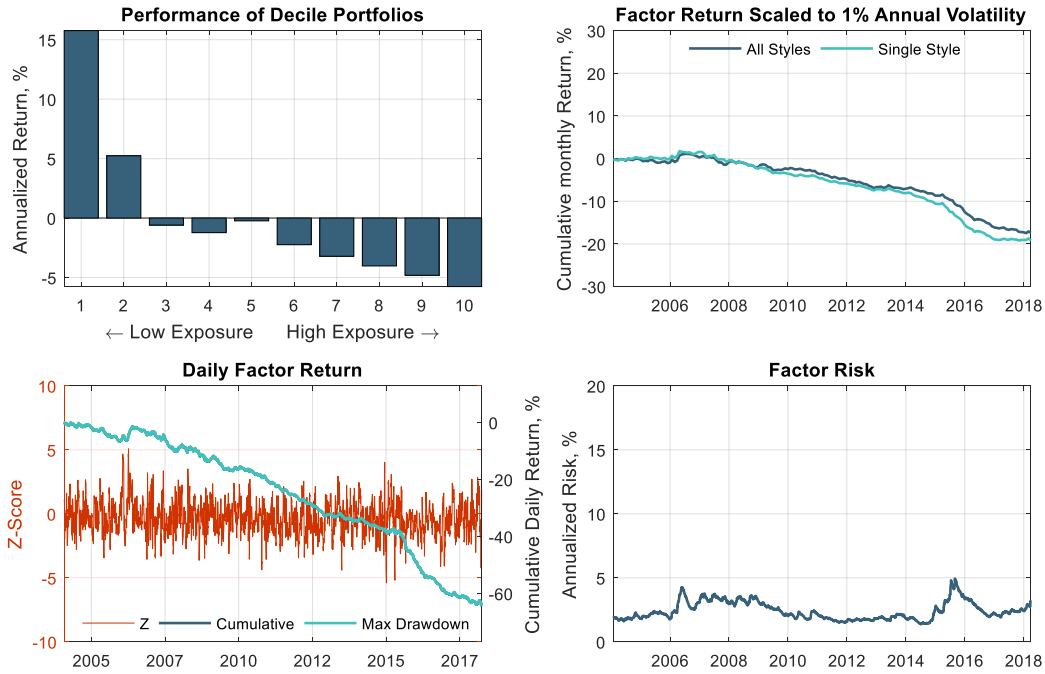


Table D.21: Summary statistics for the Mid Capitalization descriptor and factor

	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bp)	Maximum Drawdown (%)	Corr. with Market
Mid Capitalization	2.14	44.4	-3.24	2.66	-1.22	20.22	23.64	49.35	0.09

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Momentum

Description	Explains common variation in stock returns related to recent (twelve months with a lag of one month) stock price behavior. This factor is based on Systematic Equity Strategies.
Motivation	The importance of the Momentum factor in explaining stock return differences is well established in academia. The Momentum factor is one of the factors often added to the popular Fama-French Three-Factor Model. The Momentum factor phenomenon spurred a number of often-opposing models trying to explain common variation in stock returns and risk.
Start Date	Dec 31, 1998
Frequency	Daily
Exposure Interpretation	A positive exposure indicates a high medium-term momentum (good recent performance). A negative exposure indicates a low medium-term momentum (poor recent performance).
Descriptors	• Historical Alpha • Relative Strength 12-month

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Figure D.22: Deciles portfolios, performance, and risk of the Momentum factor

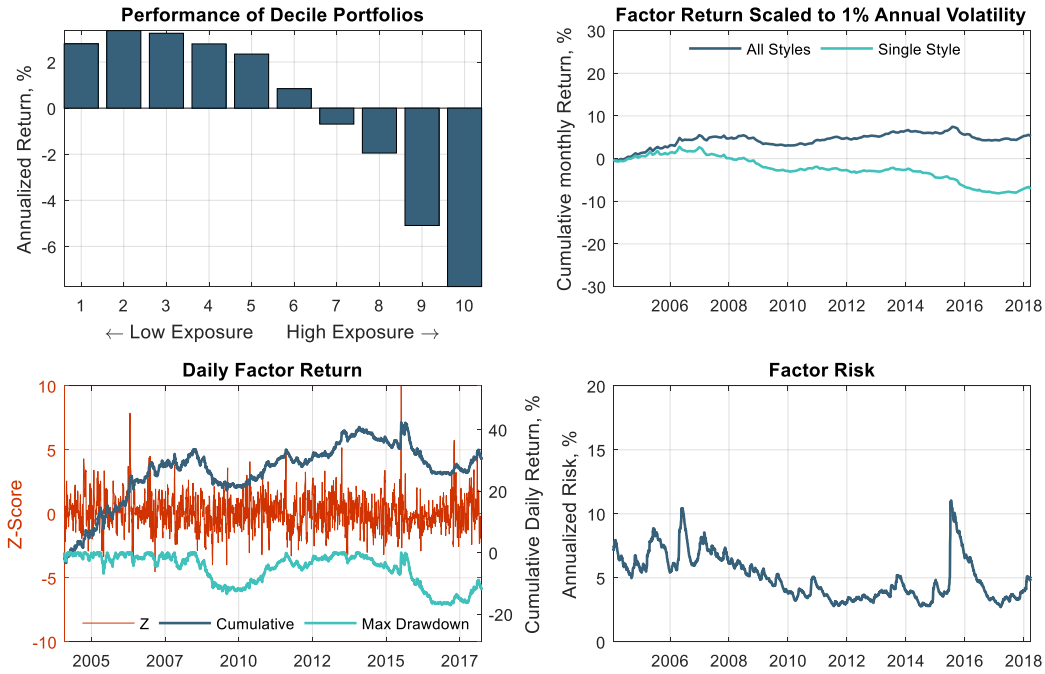


Table D.22: Summary statistics for the Momentum descriptors and factor

	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bp)	Maximum Drawdown (%)	Corr. with Market
Relative Strength 12-month	2.97	60.2	1.48	5.09	0.29	42.97	46.66	15.82	-0.14
Historical Alpha	2.71	52.0	1.90	4.43	0.43	38.82	42.50	14.88	-0.10
Momentum	2.96	57.3	1.80	4.88	0.37	44.72	48.24	15.80	-0.12

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Figure D.23: Cumulative monthly return of the Momentum descriptors

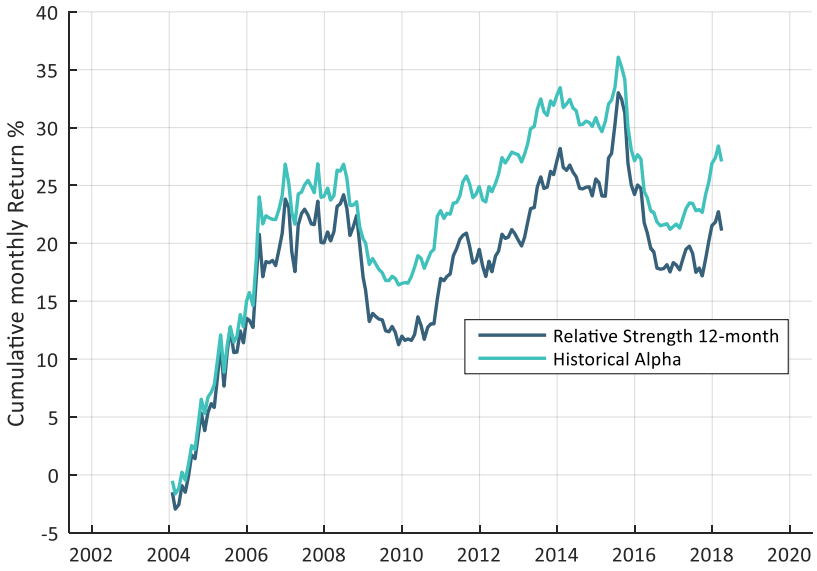


Table D.23: Correlation of the monthly returns of the Momentum descriptors and factor

	Relative Strength 12-month	Historical Alpha	Momentum
Relative Strength 12-month	1		
Historical Alpha	0.76	1	
Momentum	0.94	0.94	1

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Profitability

Description	A combination of profitability measures that characterizes efficiency of a firm's operations and total activities. This factor is based on Systematic Equity Strategies.
Motivation	From an academic point of view, the importance of profitability may be demonstrated by using the Dividend Discount Model (DDM). Under some simplifying assumptions, the Dividend Discount Model implies that higher expected future earnings imply a higher expected stock return. Following recent academic research, we use profitability measures as a proxy for future expected earnings. See Novy-Marx (2013)
Start Date	Mar 31, 2004
Frequency	Daily
Exposure Interpretation	A positive exposure indicates a high profitability and operating efficiency. A negative exposure indicates a low profitability and operating efficiency.
Descriptors	• Asset Turnover • Gross Profitability • Gross Profit Margin • Return on Assets

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Figure D.24: Decile portfolios, performance, and risk of the Profitability factor

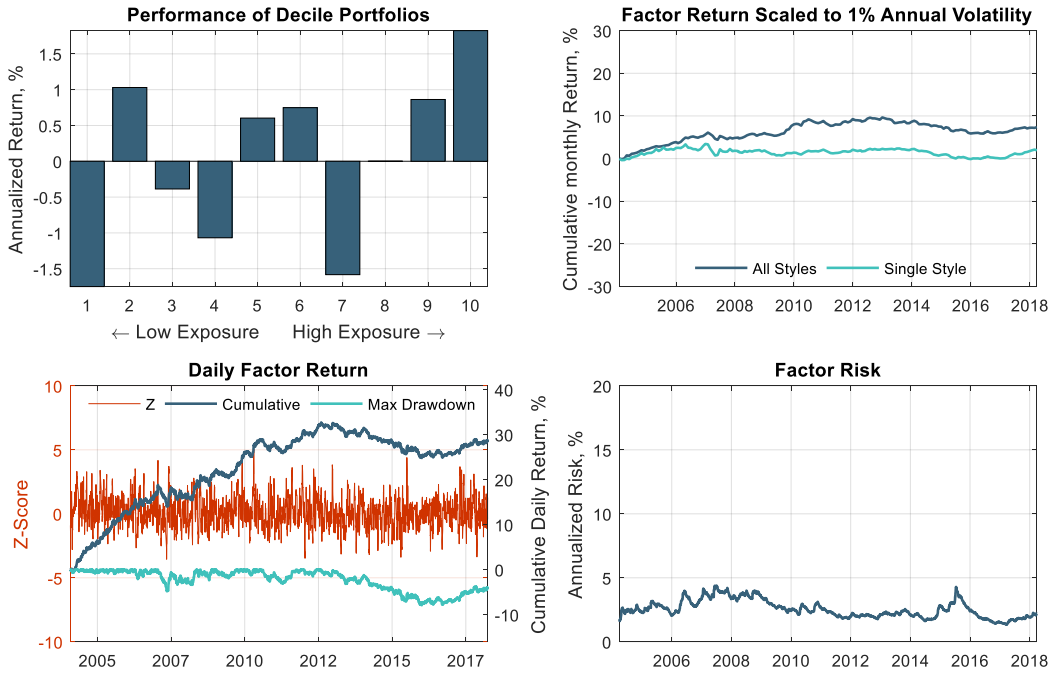


Table D.24: Summary statistics for the Profitability descriptors and factor

	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bp)	Maximum Drawdown (%)	Corr. with Market
Asset Turnover	1.14	15.6	-0.40	1.53	-0.26	-2.11	2.78	10.39	-0.13
Gross Profitability	1.24	21.6	0.99	1.95	0.51	0.32	5.31	10.18	-0.22
Gross Profit Margin	1.12	16.8	1.06	1.48	0.72	-2.26	2.66	2.46	-0.07
Return on Assets	1.29	23.4	0.35	2.47	0.14	-0.15	5.09	8.22	-0.06
Profitability	1.28	17.0	1.15	2.23	0.52	0.30	5.35	8.44	-0.23

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Figure D.25: Cumulative monthly return of the Profitability descriptors

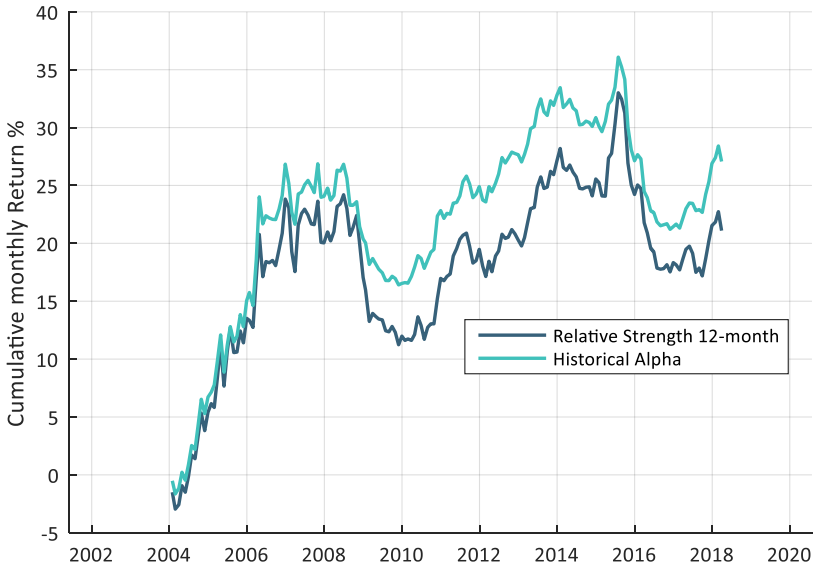


Table D.25: Correlation of the monthly returns of the Profitability descriptors and factor

	Asset Turnover	Gross Profitability	Gross Profit Margin	Return on Assets	Profitability
Asset Turnover	1				
Gross Profitability	0.46	1			
Gross Profit Margin	-0.51	0.31	1		
Return on Assets	0.19	0.58	0.34	1	
Profitability	0.46	0.9	0.45	0.81	1

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Residual Volatility

Description	Captures relative volatility in stock returns that is not explained by differences in stock sensitivities to market returns (Beta factor).
Motivation	There is persuasive evidence that stocks with high residual (idiosyncratic) volatility relative to the Capital Asset Pricing Model (CAPM) or Fama-French Three-Factor Model have unexpectedly low average returns. See, Ang *et al.* (2006) and Bali (2008) for more discussions. We include the Residual Volatility factor to capture this pervasive phenomenon.
Start Date	Dec 31, 1998
Frequency	Daily
Exposure Interpretation	A positive exposure indicates a high residual volatility. A negative exposure indicates a low residual volatility.
Descriptors	• Historical Sigma • Daily Standard Deviation • Cumulative Range

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Figure D.26: Decile portfolios, performance and risk for the Residual Volatility factor

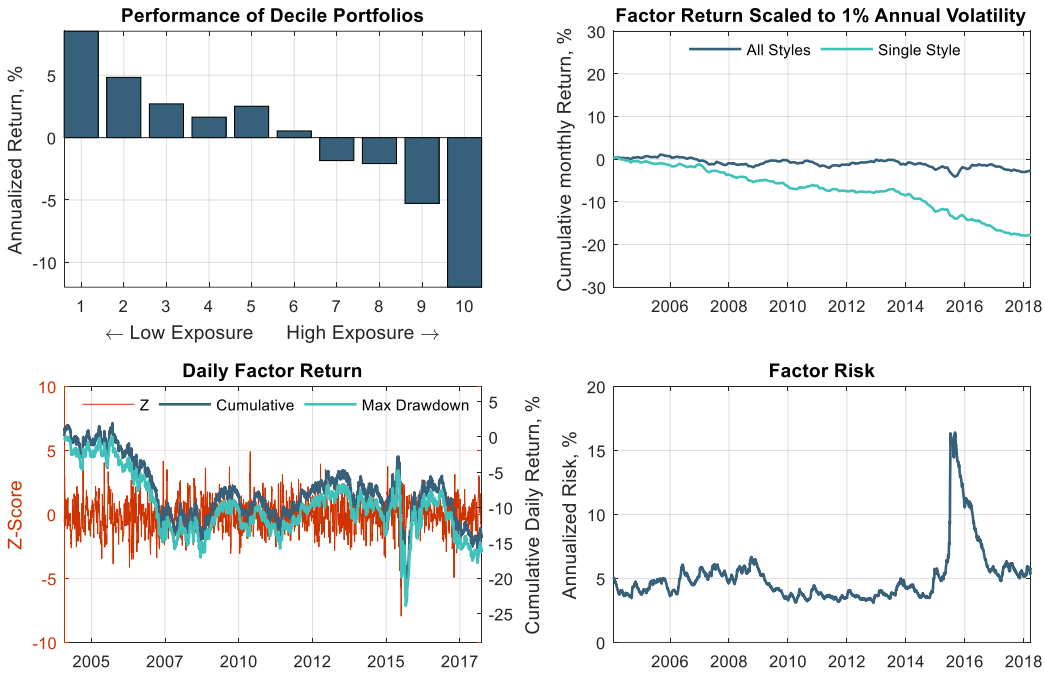


Table D.26: Summary statistics for the Residual Volatility descriptors and factor

	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bp)	Maximum Drawdown (%)	Corr. with Market
Historical Sigma	2.19	47.4	0.99	3.91	0.25	15.89	20.58	9.58	0.22
Daily Standard Deviation	2.97	55.0	-2.97	5.94	-0.50	31.67	36.40	53.48	0.38
Cumulative Range	2.12	43.9	0.67	3.94	0.17	16.87	21.77	12.16	0.16
Residual Volatility	2.75	57.3	-0.89	4.78	-0.19	28.68	33.08	24.73	0.29

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Figure D.27: Cumulative monthly return of the Residual Volatility descriptors

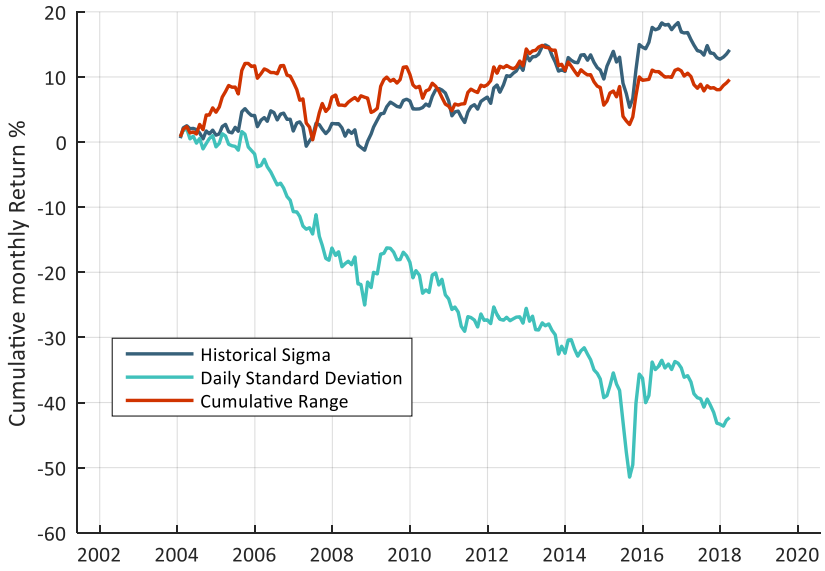


Table D.27: Correlation of the monthly returns of the Residual Volatility descriptors and factor

	Historical Sigma	Daily Standard Deviation	Cumulative Range	Residual Volatility
Historical Sigma	1			
Daily Standard Deviation	0.71	1		
Cumulative Range	0.57	0.67	1	
Residual Volatility	0.75	0.68	0.62	1

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Size

Description	Captures differences in stock returns and risk due to differences in of the market capitalization of companies.
Motivation	The importance of the Size factor in predicting the cross-section of stock returns has a long history in Barra modelling and academic literature. Also, the Size factor is one of the factors in Fama-French Three-Factor Model. There is consensus that there are significant differences in the behavior of risk and returns of large-capitalization and small-capitalization companies. Historically, small-capitalization companies earned higher returns realizing a higher volatility.
Start Date	Dec 31, 1998
Frequency	Daily
Exposure Interpretation	A positive exposure indicates large capitalization. A negative exposure indicates small capitalization.
Descriptors	• Log of Market Capitalization

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Figure D.28: Decile portfolios, performance and risk for the Size factor

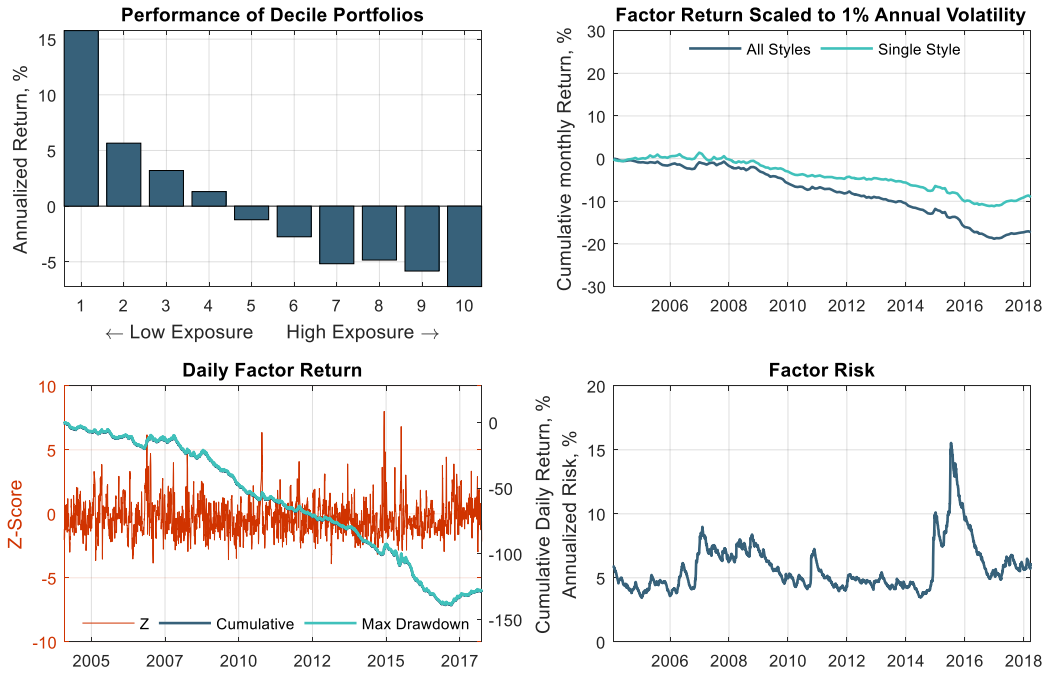


Table D.28: Summary statistics for the Size factor

	Average |t|	Percent of |t|>2	Annual Return (%)	Annual Volatility (%)	IR	CV R^2 Gain (bps)	Stud R2 Gain (bps)	Maximum Drawdown (%)	Corr. with Market
Size	4.02	70.2	-7.59	6.28	-1.21	80.38	82.55	117.74	0.01

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Appendix E: Descriptor Definitions

This appendix defines the descriptors in the style factors. The descriptors are listed under the style factors to which they belong. The factors are listed alphabetically.

Style: **Beta**

Descriptors:	HBETA	Historical Beta
		Computed as the slope coefficient from a time-series regression of stock excess returns against the cap-weighted excess returns of the estimation universe over a trailing window of 504 trading days, with a 252-day half-life. The returns are aggregated over four-day windows to reduce the effect of non-synchronicity and auto-correlation.

Style: **Book-to-Price**

Descriptors:	BTOP	Book-to-Price
		Computed by dividing the last reported book value of common equity by the current market capitalization.

Style: **Dividend Yield**

Descriptors:	DTOP	Dividend-to-Price
		Computed by dividing the trailing 12-month dividend per share by the price at the last month end.
	DTOPF	Analyst-Predicted Dividend-to-Price
		Computed by dividing the 12-month forward-looking dividend per share (DPS) by the current price.

Style: **Earnings Quality**

Descriptors:	ABS	Accruals - Balance Sheet Version
		The balance-sheet-based accruals, $ACCR_BS$, is first computed from (i) consecutive changes in balance sheet items and (ii) depreciation. The descriptor is then computed as the negative $ACCR_BS$ normalized by total assets: $$ABS = -ACCR_BS/TA$$ Thus, it goes long companies with low accruals.

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	ACF	Accruals - Cash flow Statement Version The cash-flow-statement-based operating accruals, $ACCR_CF$, is first computed from (i) items from the most recent cash flow statement and (ii) depreciation. The descriptor is then computed as the negative $ACCR_CF$ normalized by total assets: $$ACF = -ACCR_CF/TA$$ Thus, it goes long companies with low accruals.
Style:	**Earnings Variability**	
Descriptors:	VSAL	Variability in Sales Computed by dividing the standard deviation of the annual sales of the last five fiscal years by the average annual sales.
	VERN	Variability in Earnings Computed by dividing the standard deviation of the annual earnings of the last five fiscal years by the average annual earnings.
	VFLO	Variability in Cash-flows Computed by dividing the standard deviation of the annual cash flows of the last five fiscal years by the average annual cash flow.
	ETOPF_STD	Standard deviation of Analyst Forecast Earnings-to-Price Computed by dividing the standard deviation of the 12-month forward-looking earnings per share estimates by the current price.
Style:	**Earnings Yield**	
Descriptors:	CETOP	Cash-Earnings-to-Price Computed by dividing the trailing 12-month cash earnings by the current market capitalization.
	ETOP	Earnings-to-Price Computed by dividing the trailing 12-month earnings by the current market capitalization.

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	EM	Enterprise Multiple (EBIT to EV) Computed by dividing the earnings before interest and taxes (EBIT) from the last fiscal year by the current enterprise value (EV).
	ETOPF	Analyst-Predicted Earnings-to-Price Computed by dividing the 12-month forward-looking earnings by the current market capitalization.
Style:	**Growth**	
Descriptors:	EGRLF	Analyst Predicted Earnings Long-term Growth Long-term (3-5 years) earnings growth forecasted by analysts.
	EGRO	Earnings per Share Growth Rate Computed by dividing the slope coefficient from the regression of the annual earnings per share from the last five fiscal years against time, by the average annual earnings per share.
	SGRO	Sales per Share Growth Rate Computed by dividing the slope coefficient from the regression of the annual sales per share from the last five fiscal years against time, by the average annual sales per share.
Style:	**Investment Quality**	
Descriptors	AGRO	Total Assets Growth Rate Computed by first dividing the slope coefficient from the regression of the total assets from the last five fiscal years against time by the average total assets, and then multiplying by -1 to reverse the sign.
	IGRO	Issuance Growth Computed by first dividing the slope coefficient from the regression of the number of shares outstanding from the last five fiscal years against time by the average number of shares outstanding, and then multiplying by -1 to reverse the sign.
	CXGRO	Capital Expenditure Growth Computed by first dividing the slope coefficient from the regression of the capital expenditures from the last five fiscal years against time by the average capital expenditures, and then multiplying by -1 to reverse the sign.

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Style: **Leverage**

Descriptors:

MLEV — Market Leverage

Computed as,

$$MLEV = \frac{ME + PE + LD}{ME}$$

where ME is the market value of common equity on the last trading day, and PE and LD are the preferred equity and long-term debt, respectively, from the last fiscal year.

BLEV — Book Leverage

Computed as,

$$BLEV = \frac{BE + PE + LD}{BE}$$

where BE, PE, and LD are the book value of common equity, preferred equity, and long-term debt, respectively, from the last fiscal year.

DTOA — Debt-to-Assets

Computed as,

$$DTOA = \frac{TL}{TA}$$

where TL and TA are the total liabilities and total assets, respectively, from the last fiscal year.

Style: **Liquidity**

Descriptors:

STOM — Monthly Share Turnover

Computed as the log of the percentage of shares traded in the most recent month.

STOQ — Quarterly Share Turnover

Computed as the log of the average percentage of shares traded monthly over the last three months.

STOA — Annual Share Turnover

Computed as the log of the average percentage of shares traded monthly over the last 12 months.

ATVR — Annualized Traded Value Ratio

Computed as the exponentially-weighted sum of the percentage of shares traded daily over a trailing 252-day window with a 63-day

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half-life.

Style:	**Long-Term Reversal**	
Descriptors:	LTRSTR	Long-term Relative Strength
		The non-lagged Long-term Relative Strength is first computed as the exponentially-weighted sum of the log excess returns of the stock relative to the market over a trailing 1040-day window, with a 260-day half-life. The final LTRSTR descriptor is then computed by first taking the equal-weighted average of the non-lagged values over an 11-day window lagged by 273 days, and then multiplying it by -1 to reverse the sign.
	LTHALPHA	Long-term Historical Alpha
		The non-lagged Long-term Historical Alpha is first computed as the intercept term from a CAPM regression similar to the one used to compute the HBETA descriptor, except with a 1040-day window and a 260-day half-life. The final LTHALPHA descriptor is then computed by first taking the equal-weighted average of the non-lagged values over an 11-day window lagged by 273 days, and then multiplying it by -1 to reverse the sign.
Style:	**Mid Capitalization**	
Descriptors:	MIDCAP	Cube of Size Exposure
		The Size factor exposure is first cubed, and then orthogonalized to Size on a regression-weighted basis, and finally winsorized and standardized.
Style:	**Momentum**	
Descriptors:	RSTR	Relative Strength 12-month
		The non-lagged Relative Strength is first computed as the exponentially-weighted sum of the log excess returns of the stock relative to the market over a trailing 252-day window, with a 126-day half-life. The final RSTR descriptor is then computed as the equal-weighted average of the RS over an 11-day window lagged by 11 days.
	HALPHA	Historical Alpha
		The non-lagged Historical Alpha is first computed as the intercept term from the same time-series regression that is used

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to compute HBETA. The final HALPHA descriptor is then computed as the equal-weighted average of the non-lagged values over an 11-day window lagged by 11 days.

Style: **Profitability**

Descriptors:

ATO — Asset Turnover

Computed as,

$$ATO = \frac{Sales}{TA}$$

where Sales is the trailing 12-month sales, and TA is the most recently reported total assets.

GP — Gross Profitability

Computed as,

$$GP = \frac{Sales - COGS}{TA}$$

where Sales, COGS, and TA are the sales, cost of goods sold, and total assets, respectively, from the last fiscal year.

GPM — Gross Profit Margin

Computed as,

$$GPM = \frac{Sales - COGS}{Sales}$$

where Sales and COGS are the sales and cost of goods sold, respectively, from the last fiscal year.

ROA — Return on Assets

Computed as,

$$ROA = \frac{Earnings}{TA}$$

where Earnings is the trailing 12-month earnings, and TA is the most recently reported total assets.

Style: **Residual Volatility**

Descriptors:

HSIGMA — Historical Sigma

Computed as the volatility of the residual returns from the same time-series regression that is used to compute HBETA.

DASTD — Daily Standard Deviation

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		Computed as the volatility of daily excess returns over the past 252 trading days with a 42-day half-life.
	CMRA	Cumulative Range Computed as the gap between the highest and lowest points of the cumulative log excess return in the past 12 months.
Style:	**Size**	
Descriptors:	LNCAP	Log of Market Capitalization Computed as the natural logarithm of the market capitalization of the firm.

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Appendix F: Decomposing RMS Returns

We decompose excess stock returns r_n into a systematic component due to the factors, and a stock-specific component u_n. The factor returns f_k are estimated each period by a cross-sectional regression:

$$r_n = \sum_k X_{nk} f_k + u_n \,, \tag{F1}$$

where X_{nk} is the exposure of stock n to factor k. The specific returns are assumed to be uncorrelated with the factors as well as with one another.

The total R-squared (R^2) of a regression measures the cross-sectional variance explained by the factors:

$$R^2 = 1 - \frac{\sum_n w_n u_n^2}{\sum_n w_n r_n^2} \tag{F2}$$

where w_n is the regression weight of stock n. The root mean square (RMS) return, computed as:

$$RMS = \sqrt{\sum_n w_n r_n^2} \tag{F3}$$

measures the cross-sectional dispersion from zero return. As described by Menchero and Morozov (2011), the RMS return can be exactly decomposed into factor and specific return sources using a cross-sectional version of the x-sigma-rho formula,

$$RMS = \sum_k f_k \, \sigma(X_k) \rho(X_k, r) + \sigma(u) \, \rho(u, r) \tag{F4}$$

where $\sigma(X_k)$ is the RMS dispersion of factor k, and $\rho(X_k, r)$ is the cross-sectional correlation of factor k and the asset returns. The last term in Equation F4 represents the contribution to RMS from stock-specific sources.

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Appendix G: Review of Bias Statistics

G.1. Bias Statistics

A commonly-used measure for the forecast accuracy of a risk model is the bias statistic. Conceptually, the bias statistic represents the ratio of realized risk to forecast risk. Let R_{nt} be the return of portfolio n over period t, and σ_{nt} be the beginning-of-period volatility forecast. Assuming perfect forecasts, the standardized return,

$$b_{nt} = \frac{R_{nt}}{\sigma_{nt}} \tag{G1}$$

has an expected standard deviation of one. The bias statistic for portfolio n is the realized standard deviation of the standardized returns,

$$B_n = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(b_{nt} - \bar{b}_n)^2} \tag{G2}$$

where T is the number of periods in the observation window.

Assuming normally-distributed returns and perfect risk forecasts, the bias statistic B_n for a sufficiently large T is approximately normally distributed about one, and roughly 95 percent of the observations fall within the confidence interval,

$$B_n \in \left[1 - \sqrt{2/T}, 1 + \sqrt{2/T}\right] \tag{G3}$$

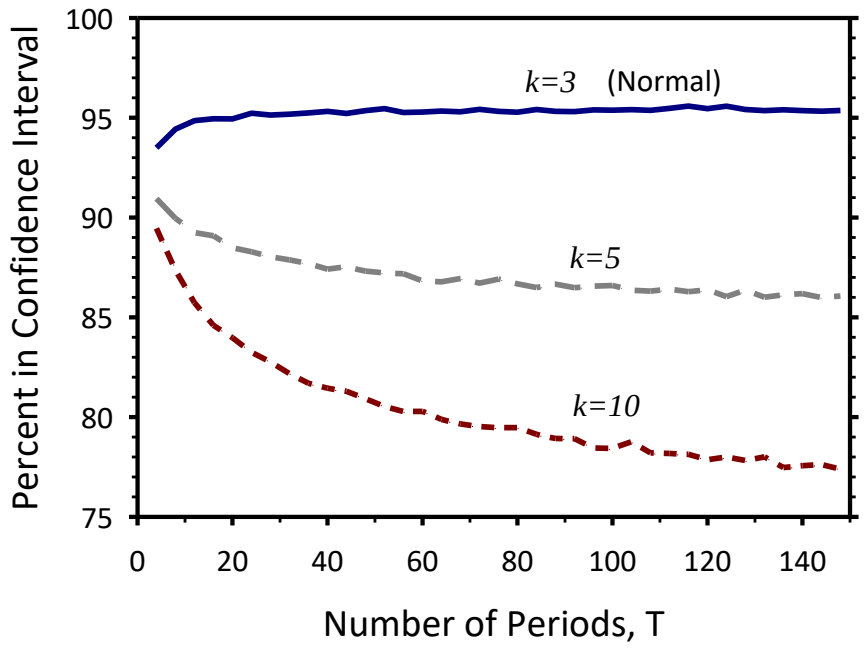
If B_n falls outside this interval, we reject the null hypothesis that the risk forecast is accurate.

If returns are not normally distributed, however, then fewer than 95 percent of the observations will fall within the confidence interval, even for perfect risk forecasts. In Figure G.1, we show simulated results for the percentage of observations actually falling within this interval, plotted versus observation window length T, for several values of kurtosis k.

For the normal case (kurtosis $k = 3$), except for the smallest values of T, the confidence interval indeed captures about 95 percent of the observations. As the kurtosis increases, however, the percentage falling within the interval drops significantly. For instance, even for a fairly modest kurtosis level of 5, only 86 percent of bias statistics fall inside the confidence interval for an observation window of 120 periods.

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Figure G.1: Percent of observations falling within the confidence interval $1 \pm \sqrt{2/T}$, where T is the number of periods in the observation window. Results are simulated using a normal distribution ($k = 3$), and using a t-distribution with kurtosis values $k = 5$ and $k = 10$. The standard deviation is equal to one in all cases. For the normal distribution, the percentage of observations inside the confidence interval quickly approaches 95 percent. As kurtosis is increased, however, the proportion within the confidence interval declines considerably.



One shortcoming of the bias statistic is that over a long period, we may have sub-periods of over-forecasting and under-forecasting, yet obtain a bias statistic close to one over the entire period. Forecasting errors can cancel out over the long term, even though the accuracy may be poor over sub-periods.

G.2. Q-Statistic

The Q-statistic is defined as:

$$Q_{nt} = b_{nt}^2 - \ln(b_{nt}^2)\,. \qquad \text{(G4)}$$

It penalizes both under and over forecast and is not prone to the type of error cancellation described above for bias statistic when averaged across time or test portfolios. We define the average Q-statistic as:

$$\bar{Q} = \frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T} Q_{nt} \qquad \text{(G5)}$$

Further information on Q-statistic can be found in Patton (2011).

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Appendix H: Covariance Matrix Estimation

Estimation of the factor covariance matrix follows a multiple-step process. First, we compute the factor correlation matrix from daily factor return series. An exponential weighting scheme, characterized by the factor correlation half-life parameter τ_ρ^F, is used to give more weight to recent observations. This is an effective method for dealing with data non-stationarity.

The Barra China A Long-Term Model has a one-month prediction horizon. The factor correlation matrix, however, is estimated from daily factor returns. We therefore must account for serial correlation in factor returns. We employ the Newey-West methodology (1987) to account for serial-correlation effects. A key parameter in this approach is the lag L_ρ^F over which serial correlation is deemed important. For instance, $L_\rho^F = 2$ implies that the return of any factor may be correlated with the return of any other factor within a two-day window.

Another complication in estimating the factor correlation matrix arises from missing factor returns. Missing factor returns may arise from using time series of differing lengths. For instance, an industry factor may start later than other industry factors in the model. We proxy the returns of this industry factor in its early history with the returns of its parent industry. Some country factors also start late, and it is not obvious how to proxy the returns of their early history. We thus use the Expectation Maximization (EM) algorithm of Dempster (1977) to estimate the correlation matrix. This method estimates the correlation matrix through an iterative procedure. It also refines the correlation forecasts as new information is incorporated into the model.

With the correlation matrix thus computed, the next step is to estimate the factor volatilities. We first estimate daily factor volatilities as exponentially-weighted averages, with a half-life of τ_σ^F. We then scale these daily volatilities first by a simple square-root-of-time factor, which is uniform for all the factors, and then by an auto-correlation correction factor estimated separately for each factor to obtain the factor volatilities for the intended horizon. The auto-correlation correction factors are estimated by the Newey-West methodology, with a lag L_σ^F and a half-life much longer than τ_σ^F to achieve stability.

Finally, we construct the covariance matrix by combining the factor correlation matrix and factor volatilities. The covariance between factors i and j is given by:

$$F_{ij} = \rho_{ij}\sigma_i\sigma_j \qquad \text{(H1)}$$

where σ_i and σ_j are the factor volatilities and ρ_{ij} is the correlation.

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Appendix I: Model Estimation Parameters

The following two tables contain factor covariance matrix and specific risk estimation parameters for the Barra China A Long-Term Model. The parameters are described by Menchero, Orr, and Wang (2011).

Table I.1: Factor covariance matrix estimation parameters. All values are reported in trading days.

Model Variant	Factor Volatility Half-life	Newey-West Volatility Lag	Factor Correlation Half-life	Newey-West Correlation Lag	Factor VRA Half-life
Responsive	84	10	504	3	21
Stable	252	10	756	5	84

Table I.2: Specific risk estimation parameters. Except for the dimensionless shrinkage parameter q, all values are reported in trading days.

Model Variant	Specific Volatility Half-life	Newey-West Auto-Correlation Lag	Newey-West Auto-Correlation Half-life	Bayesian Shrinkage Parameter q	Specific VRA Half-life
Responsive	84	5	252	0.15	21
Stable	252	5	252	0.15	84

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